

MAYO CLINIC

Mayo Clinic Research Fellows' Association (MRFA)

MRFA Research Day Florida 2026

Abstract Book

Thursday, October 29, 2026

Kinne Auditorium, Mayo Clinic in Florida, Jacksonville

Welcome

Mayo Clinic Research Fellows' Association (MRFA) Research Day, Florida 2026. Thursday, October 29, 2026, Kinne Auditorium, Mayo Clinic in Florida.

This book collects the 22 oral and rapid-fire presentations and 62 posters presented at Research Day, in program order. Posters that accompany an oral presentation point to that abstract. Abstracts appear as submitted by the authors. The live program, speaker details and Q&A are at **mrfa-florida.github.io/research-day-2026**. Contact: **MRFAFlorida@mayo.edu**.

PROGRAM AT A GLANCE

8:00 AM to 9:00 AM	Registration and breakfast
9:00 AM to 9:15 AM	Welcome and opening remarks Regina Castaneda, MD, MRFA Florida Chair; Sebastian Fernandez-Bussy, MD, Dean of Research
9:15 AM to 10:30 AM	Oral and rapid-fire presentations I Abstracts O-01 to O-11. Moderator: Regina Castaneda, MD
10:30 AM to 11:30 AM	Poster session I Posters P-01 to P-31
11:30 AM to 12:00 PM	Lunch
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1:15 PM to 2:15 PM	Poster session II Posters P-32 to P-63
2:15 PM to 2:30 PM	Awards ceremony and closing remarks Best Full Oral Presentation, Best Rapid-Fire Presentation, Best Poster Presentation and Audience Choice Award

LEADERSHIP



Regina Castaneda, MD
Chair, Mayo Clinic Research Fellows' Association (MRFA), Florida



Sebastian Fernandez-Bussy, MD
James C. and Sarah K. Kennedy Dean of Research, Mayo Clinic in Florida

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ORAL PRESENTATIONS

Session I

9:15 AM to 10:30 AM, Kinne Auditorium. Moderator: Regina Castaneda, MD

O-01 FULL ORAL Session I, talk 1 9:15 AM, 10 min Basic Sciences

Targeting P2RX7 to Suppress Innate Immune Activation Induced by Extracellular Vesicles Carrying Mitochondrial Cargo in Tauopathy

Victor Bodart-Santos, PhD, Arun Reddy Ravula, Yang You, Mohammad Abdullah, Zhi Ruan, Justice Ellison, Stephanie Radhakishun, Nibedita Basu Ray, Bridgette C. Melvin, Aishe Kurti, Zhewei Liang, Yuta Sakai, Daniel R. O'Brien, Son N. Nguyen, Scott A. Shaffer, Lindsey Fiddes, Oleg Butovsky, Wolfdieter Springer, Seiko Ikezu, Tsuneya Ikezu

Department of Neuroscience, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Tsuneya Ikezu.

Presenter: Research Associate, Department of Neuroscience

Background. Microglia play a central role in the neuroinflammatory processes underlying Alzheimer's disease (AD), potentially accelerating disease progression through the release of pro-inflammatory cytokines and secretion of extracellular vesicles (EVs). Thus, targeting molecules driving microglial EV secretion represents a promising strategy to halt tau pathology progression. P2RX7 is an interesting target due to its high expression in microglia in AD and regulation of EV secretion. We investigated the role of P2rx7 deficiency in tau pathology, EV secretion and microglia in PS19 mouse model of tauopathy.

Methods. PS19:P2rx7KO mice (9-10 months old) were evaluated using fear conditioning test and their brain sections were assessed for brain atrophy (volumetric analysis) and tau pathology (immunostaining for AT8 and Alz50). Bulk and single-cell RNA-seq (scRNA-seq) were performed on brain tissue. To visualize microglial EV secretion in vivo, P2rx7KO mice were intracranially injected with viral vectors expressing P301L-tau in neurons and mEmerald-CD9 in microglia. Brain-derived EVs were isolated and analyzed via mass spectrometry, then injected into wild-type (WT) mice to assess microglial responses by cell sorting and RNA-seq. Primary cultured neurons and microglia were used to confirm EV and P2rx7 function.

Results. PS19:P2rx7KO mice showed improved contextual and cued fear memory, reduced brain atrophy, and significantly decreased synaptic loss and tau pathology compared to age-matched PS19 mice. We identified a P2rx7-dependent gene module enriched in microglial and EV genes associated with tau pathology. scRNA-seq revealed a P2rx7-dependent microglial shift toward a pro-inflammatory phenotype with upregulated EV gene expression in PS19 mice, consisting with reduced microglial EV secretion observed in P2rx7-deficient mice expressing P301L-tau. EV proteome demonstrates that P2RX7 increases loading of tau and mitochondrial proteins into PS19 brain-derived EVs, enriched with mtDNA. Importantly, EVs secreted from mitochondrially stressed cells activate cytosolic DNA-sensing and interferon signaling in recipient microglia, contributing to inflammatory response.

Conclusions. We discovered a new mechanism associated with propagation of inflammatory response through EV-containing mitochondrial cargo in tauopathy. This microglial P2RX7–EV axis contributes to tau-associated neurodegeneration and presents a potential therapeutic target for mitigating AD progression.

Keywords: Alzheimer's disease, extracellular vesicles, microglia, microtubule-associated protein tau, mitochondria

Cite as: Bodart-Santos V, Ravula AR, You Y, et al. Targeting P2RX7 to Suppress Innate Immune Activation Induced by Extracellular Vesicles Carrying Mitochondrial Cargo in Tauopathy. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-01.

O-02 RAPID-FIRE Session I, talk 2 9:25 AM, 5 min Basic Sciences

Development of a Novel KrasFLEx-G12D Mouse Model for Kras-driven Cancer Research

Jie (Helen) Hu, MD, PhD, Muhammad Faheem, Justin R. Clark, Jiale Wang, Bin Wang, Yang Liu, Huijian Ma, Jiahui Zhao, Ying Wang, Baoan Ji, Yan Bi

Division of Gastroenterology and Hepatology, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Yan Bi.

Presenter: Research Fellow, Division of Gastroenterology and Hepatology

Introduction. Genetically engineered mouse models (GEMMs) harboring conditional mutant Kras alleles, such as LSL-KrasG12D, have been widely used to investigate tumorigenesis and therapeutic responses. However, the design of these alleles has an intrinsic limitation: in non-recombined tissues, the targeted allele blocks WT Kras expression, resulting in functional Kras heterozygosity, which may alter normal Kras function in multiple biological systems, including the immune system. Furthermore, homozygosity for the LSL-KrasG12D allele eliminates WT Kras expression, resulting in embryonic lethality and thereby increasing breeding time, cost, and animal use. To overcome these limitations, we aimed to develop a novel mouse model in which endogenous WT Kras expression is preserved prior to Cre-mediated recombination and subsequently switches to oncogenic KrasG12D expression.

Methods. We used the FLEx system with a KrasFLEx-G12D donor construct containing WT Kras exon 1 and an inverted mutant KrasG12D exon 1 flanked by nested, oppositely oriented loxP and lox2272 sites. These incompatible lox sites enable Cre-mediated inversion and stable switching between WT Kras and oncogenic KrasG12D expression. The construct was targeted to the endogenous Kras locus using CRISPR-Cas9 in C57BL/6J mice. Founder mice were screened for targeted integration by PCR-based genotyping. Functional activation of the KrasFLEx-G12D allele was evaluated by crossing these mice with pancreas-specific Pdx1-Cre mice. Mutant KrasG12D expression was detected by RT-PCR and Western blotting using a G12D-specific Kras antibody.

Results. Twenty F0 candidate pups were obtained, of which five (25%) demonstrated correct homology-directed repair (HDR)-mediated integration at the endogenous Kras locus by PCR. In KrasFLEx-G12D; Pdx1-Cre compound mice, mutant KrasG12D mRNA and protein were detected in the pancreas but not in WT or KrasFLEx-G12D littermate controls. Unlike the LSL-KrasG12D mice, the KrasFLEx-G12D mice can be bred to homozygosity, consistent with preservation of WT Kras expression.

Conclusions. We developed a novel KrasFLEx-G12D mouse model that preserves endogenous WT Kras expression prior to Cre activation and subsequently switches to oncogenic KrasG12D expression, providing a more physiologically relevant model with improved breeding efficiency and reduced animal use for Kras-driven cancer research.

Keywords: GEMMs, mouse model, Kras, pancreatic cancer, Kras-driven cancer

Cite as: Hu J, Faheem M, Clark JR, et al. Development of a Novel KrasFLEx-G12D Mouse Model for Kras-driven Cancer Research. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-02.

O-03 RAPID-FIRE Session I, talk 3 9:30 AM, 5 min Clinical Sciences

Wheelchair Propulsion Type and Musculoskeletal Burden After Spinal Cord Injury

Andres Monge Ferrin, MD, Muhammad Albaab Rehmat, Robert Pagan Rosado

Department of Physical Medicine and Rehabilitation, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Robert Pagan Rosado.

Presenter: Research Trainee, Department of Physical Medicine and Rehabilitation

Introduction. Manual wheelchair propulsion places substantial mechanical demands on the upper extremities, particularly the shoulder complex. However, existing evidence has focused primarily on shoulder-specific symptoms or pathology, and it remains unclear whether propulsion type is associated with broader musculoskeletal and pain outcomes. We compared self-reported arthritis, pain severity, and pain interference across manual wheelchair users, power/power-assist users, and non-wheelchair users with traumatic spinal cord injury.

Methods. We conducted a retrospective, cross-sectional secondary analysis of the National Spinal Cord Injury Statistical Center Spinal Cord Injury Model Systems Public Use Database. The sample included 10,521 unique adults who completed a follow-up interview from 2017 through 2021. Arthritis prevalence was compared using chi-square tests, and pain severity and interference using Kruskal-Wallis tests. Multivariable logistic, linear, and ordinal logistic regression evaluated associations after adjustment for age, sex, years since injury, neurologic level, and ASIA Impairment Scale grade.

Results. The sample included 4,104 manual users, 3,094 power/power-assist users, and 3,323 non-users. Arthritis prevalence was 22.7%, 29.4%, and 37.6%, respectively (P<.001). Unadjusted pain severity did not differ across groups (P=.53), whereas pain interference was higher among non-users than both wheelchair-using groups (P<.001). After adjustment, manual users had lower odds of arthritis than non-users (aOR, 0.76; 95% CI, 0.65–0.87; P<.001). Compared with manual users, non-users had higher pain severity (b=0.26; 95% CI, 0.09–0.44; P=.004) and pain interference (aOR, 1.26; 95% CI, 1.12–1.41; P<.001). Power/power-assist users also had higher pain severity (b=0.19; 95% CI, 0.03–0.35; P=.018), pain interference (aOR, 1.11; 95% CI, 1.00–1.22; P=.043), and odds of arthritis (aOR, 1.35; 95% CI, 1.18–1.54; P<.001) than manual users.

Conclusions. Manual wheelchair users had the lowest arthritis prevalence and, after adjustment, the lowest pain severity and pain interference, the inverse of what a shoulder-overuse hypothesis would predict. Because the outcomes were not joint-specific, these findings do not refute shoulder-specific risk from manual propulsion. Rather, manual propulsion may mark greater functional independence and activity. Residual confounding, selection, reverse causation, and the cross-sectional design preclude causal interpretation.

Keywords: Spinal cord injury; Wheelchair propulsion; Arthritis; Pain; Musculoskeletal Health

Cite as: Monge Ferrin A, Rehmat MA, Pagan Rosado R. Wheelchair Propulsion Type and Musculoskeletal Burden After Spinal Cord Injury. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-03.

O-04 FULL ORAL Session I, talk 4 9:35 AM, 10 min Translational Sciences

Single-cell spatial multi-omic mapping of glioblastoma reveals two targetable invasion programs and a margin invasion burden that predicts early progression

Paola Suarez Meade, MD, Nicholas Dove, Yi Liu, Asha Nair, Paula Schiapparelli, Aleeshba Basil, Hussien Al Asi, Ian Dryden, Emily Norton, Beatriz Fernanadez-Gil, Brandy Edenfield, Jennifer Kachergus, Heather Kileen, Ji Shi, Mieu Brooks, Vanessa Jones, Aubrey Thompson, Yan Luo, Kaisorn L. Chaichana, Honggan Qin, Z. Akkus, Craig Horbinski, Yan W. Asmann, Alfredo Quinones-Hinojosa

Department of Neurosurgery, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Alfredo Quiñones-Hinojosa (Brain Tumor Stem Cell Lab).

Presenter: Research Associate, Department of Neurosurgery

Glioblastoma (GBM) recurs from invasive cells at the infiltrated margin, the region left behind after surgery and the primary site of recurrence, yet clinical decision-making relies almost entirely on tissue from the resected core. We generated a paired single-cell spatial multi-omic atlas of matched tumor core and margin specimens (20 specimens, 10 newly diagnosed IDH-wild-type GBM patients), integrating Xenium 5,000-plex spatial transcriptomics with CosMx 64-plex spatial proteomics registered to neuropathologist-annotated histology. The atlas resolves the complete GBM ecosystem across four compartments: 7 malignant states and 16 immune, 10 brain-resident, and 7 vascular-stromal populations organized into 10 spatial niches. Tumor cores were dominated by MES-/AC-like malignant cells, monocyte-derived TAMs, and reactive vascular stroma, whereas margins shifted toward OPC-/NPC-like states embedded among neurons, oligodendrocytes, and homeostatic microglia. Applying orthogonal invasion signatures, we identified two biologically separable invasion programs that classify different tumor cells: mesenchymal invasion, predominant in the core and declining toward the margin, and neurodevelopmental invasion, sustained at the margin with marked inter-patient variability. Two niches were selectively permissive to invasion: the immune-active OPC-interface and proliferative-progenitor niches. Clinically, total margin invasion burden, irrespective of program, predicted early progression: margin invasion correlated inversely with progression-free survival (Spearman $\rho = -0.68$, $p = 0.045$), and patients above a 30% threshold progressed within ~3 months versus later progression in all patients below it (median PFS 1.2 vs 8.4 months; log-rank $p = 0.0072$), while core invasion showed no association ($p = 0.36$). The effect remained stable after individual adjustment for MGMT methylation, age, extent of resection, KPS, multifocality, and ventricular contact (HR 1.2–1.4 per 10 percentage points), none of which was itself prognostic. Margin invasion burden at diagnosis is a candidate biomarker to prioritize patients for early trial enrollment and intensified imaging surveillance, and each invasion program constitutes a regionally distinct therapeutic target for the cells that seed recurrence.

Keywords: Glioblastoma; Tumor microenvironment; Invasion; Extracellular Matrix; Spatial Transcriptomics; Multiomics

Cite as: Suarez Meade P, Dove N, Liu Y, et al. Single-cell spatial multi-omic mapping of glioblastoma reveals two targetable invasion programs and a margin invasion burden that predicts early progression. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-04.

O-05 RAPID-FIRE Session I, talk 5 9:45 AM, 5 min Clinical Sciences

Unraveling Menopause: Four Distinct Symptom Phenotypes

Jaime I. Castillo Silva, MD, Regina Castaneda, Juliana M. Kling, Ekta Kapoor, Kristin Cole, Stacey Winham, Joan Griffen, Chrisandra L. Shufelt, Stephanie S. Faubion

Presenting author underlined. Principal investigator: Stephanie S. Faubion.

Presenter: Research Fellow, Women's Health and Internal Medicine

Introduction. Menopause affects every woman who reaches midlife, yet its symptoms are usually measured as a total severity score or studied one at a time, leaving unknown how they combine within individual women. We asked whether menopause has an underlying structure of distinct, persistent symptom phenotypes that hold across hormone therapy (HT) use and menopausal stage and reproduce across care settings.

Methods. We analyzed the Menopause Rating Scale, in two independent Mayo Clinic registries, the Data Registry on Experiences of Aging, Menopause, and Sexuality (DREAMS; women seeking menopause care, n = 4,048) and the Hormones and Experiences of Aging registry (HERA; primary care patients surveyed without seeking menopause care, n = 3,655). Phenotypes were discovered by unsupervised latent class analysis in untreated postmenopausal DREAMS women (n = 1,361), tested for measurement invariance across HT use and menopausal stage, and replicated in HERA (pooled n = 7,703). One-year persistence was estimated with a continuous-time hidden Markov model (627 women, 1,376 visits), and prevalence across years since the final menstrual period with bias-corrected regression adjusted for HT use and care setting (n = 3,885).

Results. Four phenotypes emerged, Multisymptomatic, Urogenital, Psychological-dominant, and Minimal (Figure 1A). Women separated sharply (mean assignment probability 0.86; Figure 1B). Phenotypes were the same across HT use, menopausal stage and care setting (ΔBIC +263, +155 and +180 against group-specific profiles); only prevalence differed: HT users were more often Minimal (12% vs 6%), perimenopausal women more often Psychological-dominant (35% vs 14%), and women seeking care more often Urogenital-dominant (33% vs 23%). On average, 73% of women remained in the same phenotype one year later (Figure 1C). Over the 10 years after menopause, Psychological-dominant declined from 24% to 16% (−7.4 points; 95% CI −12.4 to −2.7) and Minimal rose from 9% to 15% (+6.0; 2.6 to 9.2), while the other two phenotypes were stable (Figure 1D).

Conclusion. Menopause is not a uniform experience but at least four distinct, persistent symptom phenotypes, consistent across treatment and menopausal stage, and replicated in an independent sample of women not seeking care. As subtyping has done for other heterogeneous conditions, this structure provides a foundation for mechanistic research, phenotype-specific trials and person-centered menopause care.

Cite as: Castillo Silva JI, Castaneda R, Kling JM, et al. Unraveling Menopause: Four Distinct Symptom Phenotypes. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-05.

O-06 RAPID-FIRE Session I, talk 6 9:50 AM, 5 min Translational Sciences

Combining Mesenchymal Stromal Cell-Conditioned Media with Contemporary Pharmacotherapy Modulates Hyperglycemic Kidney Tubular Injury and Geroprotective Signaling In Vitro

Maya Learmonth, Caroline Zinn, Isabella Rueda, LaTonya J. Hickson

Division of Nephrology, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: LaTonya J. Hickson.

Presenter: Graduate Student, Division of Nephrology

Diabetic kidney disease (DKD) is the leading cause of kidney failure worldwide, with no curative therapies. Cellular senescence and inflammation contribute to DKD pathogenesis. Mesenchymal stromal cell (MSC)-based therapies reduce inflammation and fibrosis in preclinical DKD models via paracrine mediated repair mechanisms; however, their interactions with contemporary DKD pharmacotherapies remain poorly understood.

We hypothesized that combining MSC-conditioned media (MSC-CM) with canagliflozin, a sodium-glucose cotransporter-2 inhibitor (SGLT2i), and/or tirzepatide, a dual GIP/GLP-1 receptor agonist, would enhance protection against DKD-associated tubular injury.

Human proximal tubular epithelial (HK-2) cells were exposed to high glucose (30 mM) and TGF-β (5 ng/mL) for 6 hours, followed by 18 hours of MSC-CM alone or combined with canagliflozin, tirzepatide, or both. Control, injury, and MSC media-control groups were included. Gene expression was assessed by qPCR for INHBA, TNC, CDKN1A/p21, CDKN2A/p16, IL6, SIRT1, and KL.

Injury markedly increased activin A (INHBA) expression, which was reduced with addition of MSC-CM, but no further reduction when combined with canagliflozin and/or tirzepatide. Geroprotective marker Klotho (KL) expression decreased with injury and remained reduced following MSC-CM treatment. In contrast, addition of tirzepatide and/or canagliflozin to MSC-CM significantly increased KL above healthy controls, suggesting enhanced geroprotective signaling. Senescence and inflammation markers (CDKN1A/p21, CDKN2A/p16, IL6), extracellular matrix remodeling marker (TNC) and geroprotective (SIRT1) markers were not consistently modulated between treatment groups (vs. controls). These findings suggest MSC-CM suppresses the injury-associated INHBA response whereas pharmacologic additions provide a distinct benefit to KL expression, showing potential for complementary pathways of tubular repair.

To assess clinical relevance, in vitro responses will be compared with outcomes from adults with CKD receiving MSC therapy through the Mayo Clinic Florida Regenerative Nephrology Compassionate Use Program, stratified by concurrent SGLT2i and incretin-based (GLP-1) therapy, to determine how standard of care pharmacotherapies modify MSC therapeutic responses.

Keywords: mesenchymal stem cells, chronic kidney disease, senescence, inflammation

Cite as: Learmonth M, Zinn C, Rueda I, Hickson LJ. Combining Mesenchymal Stromal Cell-Conditioned Media with Contemporary Pharmacotherapy Modulates Hyperglycemic Kidney Tubular Injury and Geroprotective Signaling In Vitro. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-06.

O-07 FULL ORAL Session I, talk 7 9:55 AM, 10 min Clinical Sciences

Cardiac Structural Recovery with GLP-1 Receptor Agonist Therapy in Heart Failure: A Multicenter Propensity-Matched Study

Alfredo Verastegui, MD[1], Regina Castaneda, MD[2], Diego Sepulveda, MD[3], Andres Acosta, MD, PhD[4], Demilade A. Adedinsewo, MBChB[5], Maria Daniela Hurtado Andrade, MD, PhD[3]

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Presenting author underlined. Principal investigator: Maria Daniela Hurtado Andrade, MD, PhD.

Presenter: Research Fellow, Divisions of Radiology and Surgery

Background. Glucagon-like peptide-1 receptor agonists (GLP-1 RAs), including semaglutide and tirzepatide, are widely used for weight management and have shown cardiovascular risk reduction in patients with obesity and type 2 diabetes. However, their effects on cardiac structure and function in patients with obesity and heart failure with reduced ejection fraction (HFrEF) remain poorly characterized, and it is unclear whether any benefits are driven by weight loss itself.

Methods. A propensity score-matched retrospective cohort study using Mayo Clinic data (2018–2024). Adults with obesity and HFrEF receiving semaglutide or tirzepatide for weight management for at least 12 months (GLP-1+) were matched 1:2 to HFrEF controls without GLP-1 RA exposure (GLP-1−) on age, sex, BMI, type 2 diabetes (exact), and baseline comorbidities (hypertension, atrial fibrillation, CAD, CKD, dyslipidemia, pulmonary hypertension, depression, anxiety, stroke/TIA, hypothyroidism) using nearest-neighbor matching with a 0.25 SD caliper. Echocardiographic parameters were extracted at baseline, year 1, and year 2. Longitudinal changes were evaluated using linear mixed-effects models adjusted for baseline values, comorbidities, and baseline use of guideline-directed medical therapies (ARNI, SGLT-2 inhibitors, MRAs), with Benjamini–Hochberg FDR correction. Results are reported as between-group differences (95% CI).

Results. We matched 146 GLP-1+ patients to 237 GLP-1− controls; baseline characteristics were similar (mean age 61 vs 62 years; 67.1% vs 72.6% male; BMI 34.3 ± 5.5 vs 34.9 ± 9.0 kg/m^2). At year 1, GLP-1+ patients had significantly greater total body weight loss than controls (−5.1 ± 8.6% vs −0.8 ± 7.8%; p<0.001). Compared to controls, GLP-1+ patients showed significantly greater improvements in LV function and remodeling: LV ejection fraction +4.1% (1.4 to 6.8), q=0.016; LV end-diastolic diameter −3.2 mm (−4.8 to −1.6), q=0.001; LV end-systolic diameter −4.0 mm (−6.1 to −2.0), q=0.001; and LV mass index −11.2 g/m^2 (−17.9 to −4.5), q=0.007; all surviving FDR correction. After additional adjustment for weight change, improvements in LV end-diastolic diameter (q=0.038), LV end-systolic diameter (q=0.024), and LV mass index (q=0.009) remained significant, while the LVEF difference was attenuated (q=0.115). No significant differences were observed in diastolic parameters. Exploratory 2-year analyses showed progressive remodeling with continued improvements in LVEF (+5.9% [2.5 to 9.3]) and LV mass index (−14.6 g/m^2 [−23.2 to −6.1]).

Conclusions. In patients with obesity and HFrEF, semaglutide or tirzepatide was associated with significant improvements in LV systolic function and reverse remodeling over 1–2 years versus matched controls. Structural remodeling persisted after adjustment for weight loss and background GDMT, suggesting partially weight-independent cardioprotective effects, while the LVEF improvement was attenuated after accounting for weight change.

Cite as: Verastegui A, Castaneda R, Sepulveda D, Acosta A, Adedinsewo DA, Hurtado Andrade MD. Cardiac Structural Recovery with GLP-1 Receptor Agonist Therapy in Heart Failure: A Multicenter Propensity-Matched Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-07.

O-08 RAPID-FIRE Session I, talk 8 10:05 AM, 5 min Clinical Sciences

MAP Score and Abdominal Aortic Calcium Score Capture Distinct Operative and Renal Risk in Partial Nephrectomy: A Single-Center Retrospective Study

Ekin Inal, MD, Young Erben, Nathalia Melo-Gonzalez, Neda Qosja, David Thiel, Christopher Jacobs, Houssam Farres

Division of Vascular and Endovascular Surgery, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Houssam Farres.

Presenter: Research Trainee, Division of Vascular and Endovascular Surgery

Introduction. The Mayo Adhesive Probability (MAP) score estimates adherent perinephric fat and operative difficulty before partial nephrectomy, while abdominal aortic calcium score (AACS) may reflect vascular and renal vulnerability. We examined the complementary value of these computed tomography (CT)-based measures in characterizing operative complexity and renal outcomes.

Methods. We retrospectively studied 301 patients who underwent partial nephrectomy. MAP was measured on both kidneys, with MAP ≥3 considered positive; AACS was analyzed continuously and at a cutoff of ≥1000. Operative outcomes included sticky fat and operative time. Renal outcomes included 12-month estimated glomerular filtration rate (eGFR), percent change in eGFR, new-onset chronic kidney disease (CKD) among patients with baseline eGFR ≥60 mL/min/1.73 m^2, and chronic renal pathology. Multivariable models adjusted for relevant clinical, tumor, and operative factors. An exploratory mediation analysis evaluated whether sticky fat explained the association between MAP and operative time.

Results. Mean age was 61.1±11.6 years, 64.1% were male, mean BMI was 29.9±6.0 kg/m^2, and mean baseline eGFR was 77.7±20.6 mL/min/1.73 m^2 (Table 1). AACS remained independently associated with chronic renal pathology (OR 1.17 per log-unit; 95% CI, 1.03–1.32; p=0.014), and AACS ≥1000 was associated with higher odds of chronic renal pathology (OR 2.19; 95% CI, 1.08–4.56; p=0.032). When AACS and bilateral MAP were modeled together, AACS remained associated with lower 12-month eGFR (p=0.029), while MAP was weaker (p=0.063) (Table 2). MAP was strongly associated with sticky fat (OR 3.6 per point; p<0.001) and longer operative time (+5.36 minutes per point; 95% CI, 2.43–8.29; p<0.001). After adjustment for sticky fat, the MAP effect fell by 75% and was no longer significant; sticky fat itself added 21.7 minutes (p=0.002). Mediation analysis showed a significant indirect effect through sticky fat (+5.03 minutes; p<0.001). Neither score was independently associated with new-onset CKD.

Conclusions. MAP and AACS appear to capture different preoperative risk domains. MAP mainly reflects the local operative environment and adherent fat, whereas AACS is more closely linked to chronic renal disease and postoperative renal function. Together, they may provide complementary rather than redundant information before partial nephrectomy.

Keywords: Partial nephrectomy, Mayo Adhesive Probability score, Abdominal aortic calcium score, Renal function

Cite as: Inal E, Erben Y, Melo-Gonzalez N, et al. MAP Score and Abdominal Aortic Calcium Score Capture Distinct Operative and Renal Risk in Partial Nephrectomy: A Single-Center Retrospective Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-08.

O-09 RAPID-FIRE Session I, talk 9 10:10 AM, 5 min Basic Sciences

APOE genotype modulates glial responses and myelin homeostasis during aging

Wenhui Qiao, PhD, Tadafumi C. Ikezu, Huan Sun, Yixing Chen, Yuka A. Martens, Shuwen Zhang, Yingxue Ren, Allison Wetmore, Monica Castanedes-Casey, Ligia Bastea, Lindsay B. M. Nevalainen, Whitney I. Davis, Zonghua Li, Ellen L. Bradstock, Gisela Xhafkollari, Peter Storz, Trace A. Christensen, Dennis W. Dickson, Eduardo N. Chini, Junmin Peng, Guojun Bu, Na Zhao

Department of Neuroscience, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Na Zhao.

Presenter: Research Associate, Department of Neuroscience

Introduction. Aging is marked by aberrant neuroinflammation driven by microglia and astrocytes, yet how apolipoprotein E (APOE) genotype, the strongest genetic modifier of longevity and Alzheimer's disease, influence glial aging remains unclear.

Methods. We used young (3 months) and aged (24 months) humanized APOE2, APOE3, and APOE4 mice to profile transcriptomic changes in microglia and astrocytes using single-cell RNA sequencing (n = 4 mice/genotype/age). In parallel, $CD11b^+$ microglia and $ACSA2^+$ astrocytes were sequentially isolated from an independent cohort (n = 6 mice per/genotype/age) for proteomic analysis to assess age-related translational changes. Key findings were validated using immunostaining and electron microscopy in additional independent cohorts.

Results. Single-cell RNA sequencing revealed the most transcriptional alterations in APOE2 across both glial types. In contrast, proteomic analyses showed the fewest changes in APOE2 and the most in APOE4, with immune-related pathways commonly altered. Validation confirmed that age-related glial activation was enriched in white matter, where aged APOE2 mice exhibited preserved myelin homeostasis, while APOE3 and APOE4 mice showed impaired myelin integrity. Aging also led to an accumulation of $CD8^+$ T cells in white matter, likely driven by activated glia rather than APOE genotype.

Conclusions. These findings highlight roles for APOE and aging in regulating glial activation, myelin integrity, and immune responses.

Keywords: Aging; APOE; Microglia; Astrocyte; Multi-omics

Cite as: Qiao W, Ikezu TC, Sun H, et al. APOE genotype modulates glial responses and myelin homeostasis during aging. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-09.

O-10 FULL ORAL Session I, talk 10 10:15 AM, 10 min Clinical Sciences

Adverse Childhood Experiences and Functional Outcomes Following Interdisciplinary Pain Rehabilitation: A Retrospective Cohort Study

Leonardo Ospino, Fiorela Martinez, MD, Sebastian Encalada, Jordan Hoprich, Johanna Mosquera-Moscoso, Christopher Sletten

Department of Pain Medicine, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Christine Hunt.

Presenter: Research Trainee, Department of Pain Medicine

Introduction. Chronic pain is influenced by biological, psychological, and social factors, including adverse childhood experiences (ACEs). Although higher ACE exposure is associated with greater psychological distress and poorer functioning, its impact on outcomes after interdisciplinary pain rehabilitation is not well understood. This study examined whether ACE burden affected treatment response and clinically meaningful improvement following participation in an interdisciplinary pain rehabilitation program.

Materials and methods. This retrospective cohort study included 985 adults with chronic pain who completed a 2-week interdisciplinary Pain Rehabilitation Center program at Mayo Clinic Florida between January 2022 and February 2025 (IRB #24-013423). Participants were categorized into high (≥3) and low (≤2) ACE groups. Outcomes included depressive symptoms (CES-D), pain catastrophizing (PCS), and health-related quality of life (SF-36), assessed at admission and discharge. Clinically meaningful improvement was defined using established MCID thresholds. Multivariable regression and mediation analyses were performed.

Results. Participants with high ACE exposure had significantly worse baseline psychological and functional status. Despite this, both groups demonstrated significant improvements across all outcomes, with mean changes meeting or exceeding MCID thresholds. Compared with the low-ACE group, high-ACE participants experienced greater reductions in depressive symptoms (−13.4 vs. −10.7, $p<0.001$) and pain catastrophizing (−11.5 vs. −9.7, $p=0.02$), as well as greater improvement in physical functioning ($p=0.006$). High ACE status independently predicted clinically meaningful improvement in depression (OR 1.82, 95% CI 1.39-2.39, $p<0.001$). Reductions in pain catastrophizing partially mediated improvements in depressive symptoms.

Discussion. Individuals with greater childhood adversity entered treatment with more severe impairment but achieved comparable or better outcomes following interdisciplinary pain rehabilitation. Trauma exposure should not be considered a barrier to rehabilitation success. Trauma-informed, biopsychosocial interventions may further enhance outcomes in chronic pain care.

Keywords: Childhood Trauma, Interdisciplinary Pain Rehabilitation, Health-Related Quality of Life, Psychological Distress, Trauma-Informed Care

Cite as: Ospino L, Martinez F, Encalada S, Hoprich J, Mosquera-Moscoso J, Sletten C. Adverse Childhood Experiences and Functional Outcomes Following Interdisciplinary Pain Rehabilitation: A Retrospective Cohort Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-10.

O-11 RAPID-FIRE Session I, talk 11 10:25 AM, 5 min Translational Sciences

Prolonged Vorasidenib Treatment Accelerates Growth in a Subset of High-Grade IDH-Mutant Glioma Cultures

Thomas Sears, PhD, Andrew Li, Craig M. Horbinski

Department of Laboratory Medicine and Pathology, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Craig Horbinski.

Presenter: Research Associate, Department of Laboratory Medicine and Pathology

Background. Gliomas bearing IDH1/2 mutations (IDHmut) represent a unique subtype of this disease that progresses from low- to high-grade lesions. Despite their relatively positive prognosis compared to IDH1/2 wildtype glioblastomas, these brain tumors are nonetheless invariably fatal. New targeted therapies that inhibit IDHmut enzyme function have emerged for low-grade gliomas, but these therapies have not yet proven to be effective in high-grade counterparts.

Methods. Five high-grade IDHmut glioma cultures were treated with the IDHmut inhibitor vorasidenib using short-term (5 days) and long-term (4 months) treatment. For long-term treatments, longitudinal sample acquisition occurred every month. Cytotoxicity (trypan blue) and proliferation (EdU) assays were conducted at each timepoint, in addition to acquisition of flash-frozen cell pellets for multi-dimensional transcriptomic, DNA methylation, and proteomic analyses.

Results. Short-term (5 day) dose-response analysis of vorasidenib between 0-1 μM showed no effect on cytotoxicity or cell proliferation in our IDHmut glioma cultures as determined by Countess cell growth and cytotoxicity assays. ELISAs showed that vorasidenib elicited maximum inhibition of D2HG production in all our cell cultures at a concentration as low as 10 nM, and that this inhibition extended through our entire 4-month treatment period. Interestingly, two different IDHmut glioma cultures (TS603, MGG152) displayed accelerated growth after 2-3 months of treatment with vorasidenib, supporting a delayed epigenetic mechanism of growth activation. Multi-dimensional transcriptomic, DNA methylation, and proteomic analyses are ongoing.

Conclusions. Considering that IDHmut create epigenomic DNA hypermethylation (G-CIMP high) through production of the oncometabolite D2HG, and that this hypermethylation is partially lost due to reduced D2HG levels during malignant tumor progression (G-CIMP low), we hypothesize that IDHmut enzyme inhibition imposes undesirable tumor growth activation in a subset of these patients. Future goals will be to correlate this work with clinical samples from vorasidenib-treated patients to identify pre- and post-treatment biomarkers that may be able to predict negative treatment response in patients receiving IDHmut inhibitors.

Keywords: Glioma; Neuro-oncology; Cancer epigenetics; Pharmacology

Cite as: Sears T, Li A, Horbinski CM. Prolonged Vorasidenib Treatment Accelerates Growth in a Subset of High-Grade IDH-Mutant Glioma Cultures. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-11.

ORAL PRESENTATIONS

Session II

12:00 PM to 1:15 PM, Kinne Auditorium. Moderator: Regina Castaneda, MD

O-12 FULL ORAL Session II, talk 1 12:00 PM, 10 min Translational Sciences

Therapeutic potential of antisense oligonucleotide targeting of Tau Tubulin Kinase 1 in models of Alzheimer's disease

Grace Samtani, PhD, Justice Ellison, Jasmine Newington, Alejandro Preciado, Mariam Stein, Prakruthi Vadakattu, Christine Hong, Fredrik Kamme, Tsuneya Ikezu, Seiko Ikezu

Department of Neuroscience, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Seiko Ikezu.

Presenter: Research Fellow, Department of Neuroscience

Background. Alzheimer's disease (AD) is a progressive neurodegenerative disorder featuring a pathological hallmark of neurofibrillary tangles composed of abnormally phosphorylated tau, which significantly contribute to synaptic degradation and cognitive dysfunction. Tau tubulin kinase-1 (TTBK1) is a brain-specific serine/threonine and tyrosine kinase implicated in tau pathology, evidenced by reports of high neuronal expression in the entorhinal cortex and hippocampal CA1 of human AD brains. We have identified several TTBK1 phosphorylation sites that overlap with PHF-tau sites (pTyr197, pSer198, pSer199, pSer202) on full-length human tau (PMID 16923168). Further, TTBK1 directly phosphorylates tau at a pre-tangle marker site (pSer422), as well as one of the AT8 recognition sites (pSer202/pThr205), illustrating its potential role in the modulation of tau pathology in AD.

Methods. Here, we employed antisense-oligonucleotide (ASO) technology to specifically target mouse Ttbk1 mRNA in PS19 (TauP301S) tauopathy mice at 6 months of age and evaluated phosphorylated tau (p-tau) levels by ELISA (PMID 37853497), hippocampal AT8+ p-tau accumulation, gliosis and synaptic density via immunofluorescence, neuroinflammation via Olink, spatial transcriptomics via MerSCOPE, and functional phenotypes via the open field test at 8-weeks post-injection.

Results. We observed a significant reduction of AT8+ p-tau in the hippocampus of ASO-treated mice, accompanied by synaptic rescue in the CA1, reduction in cortical atrophy, and reduction of PS19 mouse hypermobility phenotype. ASO administration did not augment neuroinflammation via gliosis in PS19 mice or cytokine increases in WT mice, establishing its safety and tolerability. Spatial transcriptomics revealed a decrease in p-tau-associated OPC-Oligodendrocyte cell population after ASO-Ttbk1 treatment, indicating a new cellular response for disease amelioration.

Conclusions. Together, our study establishes TTBK1 as a disease-modifying driver of tau pathology and support ASO-mediated TTBK1 suppression as a safe, translational therapeutic strategy for AD patients.

Keywords: Alzheimer's; therapeutic; tau; pathology; neuroscience

Cite as: Samtani G, Ellison J, Newington J, et al. Therapeutic potential of antisense oligonucleotide targeting of Tau Tubulin Kinase 1 in models of Alzheimer's disease. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-12.

O-13 RAPID-FIRE Session II, talk 2 12:10 PM, 5 min Clinical Sciences

What Patients Want: Are Weight-Loss Goals Realistic Relative to the Strategies Patients Are Willing to Adopt?

Diego Sepulveda, MD, Alexia R. Maier, Pamela K. Bennett, Regina Castaneda, Tina Taaffe, Alvaro Cornejo Cobo, Kirsten E. Frederiksen, Andres Acosta, Maria D. Hurtado Andrade

Division of Endocrinology, Diabetes, and Metabolism, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Maria D. Hurtado Andrade.

Presenter: Research Trainee, Division of Endocrinology, Diabetes, and Metabolism

Background. Patients seeking weight management often hold weight-loss expectations exceeding what available interventions offer. How expectations align with patients' willingness to pursue evidence-based treatments remains poorly described.

Objective. To describe weight-loss expectations, treatment preferences, and the alignment between expectations and treatment uptake among patients seeking weight management.

Methods. We retrospectively reviewed electronic medical records of adults attending an initial weight management consultation from 2019-2025. Demographics, clinical characteristics, weight-loss goals, and interest in and uptake of dietary support, behavior therapy, obesity medication (OM), endoscopic sleeve gastroplasty (ESG), and metabolic bariatric surgery (MBS) were extracted. Analyses included Student's t-tests, Pearson correlations, and ANOVA.

Results. We included 774 participants (age 51±14 years, 76% female, 91% White, weight 116±27 kg, BMI 41±8 kg/m^2); 52% had class III obesity, and 99.5% had ≥1 obesity-related disease. Of 656 (85%) interested in dietary support, 461 (62%) attended ≥1 dietitian visit; of 355 (46%) interested in behavior modification therapy, 179 (24%) attended ≥1 therapy session. Among 750 OM-eligible patients, 571 (74%) expressed interest and were prescribed one including semaglutide (32%), tirzepatide (21%), or phentermine/topiramate (19%). Among ESG- or MBS-eligible patients, 80/383 (21%) and 150/526 (29%) expressed interest, of whom 41 (51%) accepted gastroenterology referral and 89 (59%) surgical referral, respectively. Patients interested in MBS reported higher weight-loss goals than those interested in OM alone or OM+ESG (36±10% vs. 26±10% and 22±7%; p<0.001). Those not interested in any intervention reported goals of 26±12%. One-third of patients reported ≥1 non-weight goal including improvement in health (21.6%), mobility (7.7%), pain (4.8%), and appearance (2.7%).

Conclusions. Patients seeking weight management often set weight-loss goals exceeding realistic outcomes. Despite high initial interest, treatment uptake was substantially lower across all modalities, revealing a gap between interest and engagement. These findings highlight the need to calibrate patient expectations and address barriers to care uptake.

Keywords: Obesity, Weight management, GLP-1s, Bariatric Surgery

Cite as: Sepulveda D, Maier AR, Bennett PK, et al. What Patients Want: Are Weight-Loss Goals Realistic Relative to the Strategies Patients Are Willing to Adopt?. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-13.

O-14 RAPID-FIRE Session II, talk 3 12:15 PM, 5 min Translational Sciences

Metabolic modulations of CAR T cells to improve anti-tumor efficacy in Glioblastoma

Srabasti Sengupta, PhD, MS, Adriana Yamaguchi, Anna Nulk, Emma Redmon, Shahin Hamzeh, Trevor Ruiz, Nagheme Thomas, Maryam Rahman, Duane A. Mitchell, Jianping Huang, Loic P. Deleyrolle

Department of Molecular Medicine, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Loic Deleyrolle.

Presenter: Research Fellow, Department of Molecular Medicine

Introduction. Glioblastoma (GBM) is a highly aggressive and treatment-resistant brain malignancy, and the immunosuppressive tumor microenvironment (TME) substantially limits the efficacy of the chimeric antigen receptor (CAR) T-cell therapy. A major barrier is metabolic competition within the TME, where highly glycolytic GBM cells consume major glucose and generate nutrient-deprived TME that compromises the T-cell persistence and function. We hypothesize that metabolically reprogramming CAR T-cells to sustain glucose utilization will enhance their fitness and anti-tumor activity in GBM.

Methods. CD70 targeted CAR T-cells were genetically engineered to overexpress either glucose transporter- GLUT1 or GLUT3. Metabolic fitness was assessed by LiCellMo-metabolic profiling. Functional activity was evaluated through coculture with GBM cell line under glucose restricted conditions. CAR T-cell activation, target cell avidity, proliferation and cytotoxicity were assessed using ELISA, Cell avidity assay, Flowcytometry analysis and NanoLive live cell imaging. Therapeutic efficacy was further evaluated in immunocompromised orthotopic GBM mouse models by measuring intratumoral CAR T cell presence, tumor burden and overall survival.

Results. Overexpression of GLUT1/3 significantly increased glucose uptake by CD70 CAR T-cells at baseline and following antigen stimulation, demonstrating enhanced metabolic fitness. Metabolically engineered CAR T cells exhibited greater activation and functional capacity compared with control CAR t cells, including increased avidity and increased tumor cell killing specially under glucose restricted conditions that mimic the GBM TME. In vivo, GLUT1/3 overexpressing CAR T-cells demonstrated improved tumor infiltration and persistence within GBM lesions, showing enhanced tumor control and prolonged survival.

Conclusions. Metabolic enhancement of CD70 CAR T-cells through GLUT1/3 overexpression improves glucose utilization, sustains effector function under nutrient stress, and enhances antitumor activity against GBM. These findings demonstrate that immune metabolic reprogramming can overcome key metabolic barriers imposed by GBM TME and support further development of metabolically optimized CAR T-cell therapies as a strategy to improve outcomes for patients with glioblastoma.

Keywords: Glioblastoma, Immunotherapy, Metabolism, CAR-T cells, Solid tumors

Cite as: Sengupta S, Yamaguchi A, Nulk A, et al. Metabolic modulations of CAR T cells to improve anti-tumor efficacy in Glioblastoma. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-14.

O-15 FULL ORAL Session II, talk 4 12:20 PM, 10 min Clinical Sciences

Optimal window for lactic acid clearance after Impella 5.5 placement in patients with cardiogenic shock

Andrea Muniz, MD, Aarti Desai, MBBS, Rebecca Klingbeil, Hans Mautong, Parag Patel, Anna Shapiro, Rohan Goswami

Division of Heart Failure and Transplantation, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Rohan Goswami.

Presenter: Visiting Research Fellow, Cardiac Transplant

Background. Lactate clearance is a key biomarker of end-organ perfusion and prognosis in cardiogenic shock, but the optimal assessment window for prognostication after Impella 5.5 implantation remains uncertain. Current studies have shown 24-hour lactic acid levels to haev prognostic value. We aim to re-assess this optimal window of lactate clearance and take this further to show long-term trends by final outcome groups.

Methods. We retrospectively evaluated lactate trajectories in cardiogenic shock patients supported with Impella 5.5 between January 2020 and September 2025. We stratified outcomes as those transplanted, recovered, underwent durable LVAD (dLVAD) implantation, or died while supported by Impella 5.5. For survival analysis, transplant, recovery, and dLVAD were grouped as "survivors" and compared with those who died. Logistic regression was used to evaluate mortality risk.

Results. 190 patients were included: 153 transplanted, 10 recovered, 13 received dLVAD, and 14 died. Time from admission to Impella implantation and pre-Impella lactate levels did not differ across outcome groups. Pre-Impella lactate levels were not statistically different across the 4 outcome groups.

At 24 hours, lactate levels declined across all outcome groups with minimal separation between survivors and non-survivors. By 72-hours, divergence was evident: survivors demonstrated sustained clearance, while non-survivors had persistent elevation. On univariate analysis, 24-hour lactate was associated with an odds ratio (OR) 1.69 ($p=0.11$) for mortality, while 72-hour lactate levels showed a stronger association OR 6.04 ($p<0.001$). On multivariate analysis, 24-hour lactate remained significantly associated with mortality (OR 2.08, $p=0.036$), and 72-hour lactate demonstrated an even stronger association (OR 15.9, $p=0.003$).

Conclusions. These findings highlight that 72-hour lactate clearance, not 24-hour, more accurately predicts mortality after Impella 5.5 implantation.

Keywords: lactate, lactic acid clearance, cardiogenic shock, Impella 5.5, heart failure, mechanical circulatory support

Cite as: Muniz A, Desai A, Klingbeil R, et al. Optimal window for lactic acid clearance after Impella 5.5 placement in patients with cardiogenic shock. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-15.

O-16 RAPID-FIRE Session II, talk 5 12:30 PM, 5 min Translational Sciences

Information That Sparks Dialogue: Supporting Clinical Trial Communication Through an AI-Enabled Portal

Fan Wang, PhD, Andrew H. Maxim, Courtny Franco, Rashi Ghosh, Christopher You, Trevor Kauer, Stephen Anton, Elizabeth Flood-Grady, Trudy Gaillard, Thomas J. George, Benjamin Lok, Carolyn Tucker, Melissa Vilaro, Tuo Lin, Fern Webb, Diana Wilkie, Danyell Wilson-Howard, Olveen Carrasquillo, Janice L. Krieger

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Presenting author underlined. Principal investigator: Janice L. Krieger.

Presenter: Research Fellow, Department of Quantitative Health Sciences

Introduction. Facilitating patients' ability to ask informed questions about clinical trials is a component of ethical recruitment and informed consent. Clear and accessible trial information can help patients and families understand complex concepts, prepare for communication with research teams, and build confidence to ask relevant questions. ALEX (Agents Leveraging Empathy for EXams) is an AI-enabled cancer clinical trial portal. We examined what questions participants identified during ALEX, their plans for follow-up, and whether participants could find ALEX easy to understand while also wanting additional communication.

Methods. We used a mixed-methods analysis of data from 136 adults across four sites (Mayo Clinic Jacksonville/UF Gainesville/UF Jacksonville/University of Miami). From November 2023 to April 2025, we conducted 34 focus group or think-aloud sessions. Participants completed pre/post questionnaires. Transcripts were coded by thought unit, a discrete participant utterance connecting a question or information need to a next step. Spearman correlations examined one ease-of-understanding item and three motivation items, with Holm adjustment.

Results. Perceived ease of understanding was positively associated with motivation to learn more, discuss clinical research with a doctor, and discuss with people who help make health care decisions (ρ=.308-.402, all Holm-adjusted $p<.001$). 89.3% found ALEX easy to understand, 87.8% wanted to learn more, 75.6% wanted to talk to a doctor, and 74.8% wanted to talk to those who help make health care decisions. Across 22 sessions, 34 thought units were identified. Question preparation was the most frequent communication strategy (n=12), followed by clarification (n=10). Participants identified questions about source credibility, personal relevance, study requirements, and safety. For follow-up, health care providers were identified in 13 thought units and research staff in 11. Participants described ALEX as useful for helping them formulate questions and identify sources of information.

Conclusions. Participants found ALEX easy to understand and reported motivation for further information seeking about clinical trials. Qualitative findings suggest that ALEX may support patients and families in preparing questions and identifying sources for follow-up. By facilitating question preparation and communication, ALEX may enhance the quality of patient-provider communication around cancer clinical trials.

Keywords: Cancer Clinical Trials; Health Communication; Artificial Intelligence; Patient Education; Patient-Provider Communication

Cite as: Wang F, Maxim AH, Franco C, et al. Information That Sparks Dialogue: Supporting Clinical Trial Communication Through an AI-Enabled Portal. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-16.

O-17 RAPID-FIRE Session II, talk 6 12:35 PM, 5 min Basic Sciences

Humanized MAPT and APP Increase Neuronal Excitability and Accelerate Phosphorylated Tau Propagation in a Novel Entorhinal Cortex Layer II–Targeted Mouse Model of Early Alzheimer's Disease

Ayaka Tatsumoto, PhD, Arun Reddy Ravula, Brendan Gibbs, Nibedita Basu Ray, Justice Ellison, Alejandro Longtree-Preciado, Seiko Ikezu, Tsuneya Ikezu

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Presenting author underlined. Principal investigator: Seiko Ikezu.

Presenter: Research Fellow, Department of Neuroscience

Introduction. In Alzheimer's disease (AD), phosphorylated tau first appears in entorhinal cortex layer II (ECII) and subsequently spreads to the hippocampus and the cerebral cortex. We developed a mouse model that recapitulates early tau propagation by injecting AAV2/6 expressing Cre-inducible P301L human tau (AAV-FLEX-tauP301L) into ECII of Wolframin1-Cre (Wfs1) mice. In this study, to determine how human tau and amyloid-β influence tau propagation and neuronal excitability, we generated human MAPT knock-in Wfs1 (T-Wfs1) and APPNL-G-F/MAPT double-knock-in Wfs1 (AT-Wfs1) mice.

Methods. Wfs1, T-Wfs1, and AT-Wfs1 mice received unilateral ECII injections of AAV-FLEX-tauP301L at 7 months of age and were analyzed 3 months later. Immunohistochemistry assessed human tau (HT7), phosphorylated tau (AT8), misfolded tau (Alz50), Parvalbumin/Somatostatin positive inhibitory (PV+/SST+ IN) and Neurogranin positive excitatory neurons (NRGN+ EN). Field electrophysiology measured spontaneous activity in hippocampal and neocortical regions before and after GABA-A receptor blockade with gabazine. GeoMx Digital Spatial Profiling was performed on PV+IN and NRGN+EN from uninjected 7-month-old Wfs1 and AT-Wfs1 brains.

Results. HT7+ tau reached CA1 and subiculum in Wfs1 mice, whereas AT8+ and Alz50+ tau were absent. In T-Wfs1 mice, HT7+, AT8+, and Alz50+ tau extended to CA1, subiculum, and neocortex and were further increased in AT-Wfs1 mice. Of total AT8+ tau, 34.7% colocalized with NRGN+ EN, whereas <1% colocalized with PV+/SST+ IN, indicating preferential propagation to excitatory neurons. T-Wfs1 and AT-Wfs1 mice showed increased spontaneous firing compared with Wfs1 mice, with little further increase after gabazine, consistent with impaired GABAergic transmission. Both lines also exhibited reduced PV+ IN numbers independent of tau spread. In PV+ IN from AT-Wfs1 mice, genes related to fast-spiking, Ca2+ homeostasis, excitatory recruitment, and synaptic function were selectively downregulated, together with a predicted upstream transcriptional regulator.

Conclusions. Human tau and amyloid pathology cooperatively enhance neuronal hyperexcitability and tau propagation from ECII to hippocampal and neocortical regions, implicating inhibitory circuit dysfunction in early AD. Ongoing studies test whether recipient-cell excitability regulates tau propagation using DREADDs and whether transcriptional regulator agonism can restore E/I balance in AT-Wfs1 mice.

Keywords: Alzheimer's Disease, Tau Propagation, Braak Staging, Humanized Animal Models, Neuronal Excitability

Cite as: Tatsumoto A, Ravula AR, Gibbs B, et al. Humanized MAPT and APP Increase Neuronal Excitability and Accelerate Phosphorylated Tau Propagation in a Novel Entorhinal Cortex Layer II–Targeted Mouse Model of Early Alzheimer's Disease. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-17.

O-18 FULL ORAL Session II, talk 7 12:40 PM, 10 min Clinical Sciences

Early Revascularization in Symptomatic Carotid Artery Disease Leads to Better Outcomes: A Ten-Year Multi-Center Cohort Analysis

Diego Sanchez-Calderin, MD, James F. Meschia, Gyanendra Kumar, Robert D. Brown Jr, Bernard R. Bendok, Kevin Barrett, W. Chris Fox, David A. Miller, Camilo Polania-Sandoval, Ekin Inal, Camila Esquetini-Vernon, Sukhwinder S. J. Sandhu, Richard D. Beegle, Houssam Farres, Giuseppe Lanzino, Young Erben

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Presenting author underlined. Principal investigator: Young Erben.

Presenter: Research Trainee, Division of Vascular and Endovascular Surgery

Introduction. Early intervention for symptomatic carotid artery disease (CAD) may benefit patients, yet the optimal timing window is debated. This study evaluated the impact of timing of revascularization on perioperative and long-term outcomes.

Methods. We reviewed 821 carotid revascularization procedures for CAD from 2016-2026. Procedures were stratified by into urgent (0–2 days), early (3–14 days), and delayed (≥15 days) groups. Endpoints included 30-day and long-term mortality, ipsilateral stroke, transient ischemic attack (TIA), myocardial infarction (MI), and a composite of these events, plus restenosis and reintervention. Cox regressions identified independent predictors of the outcomes. Sensitivity analyses adjusted for factors related to adjunct procedures at the time of initial neurovascular event presentation.

Results. The cohort's median age was 74.2 years; the majority were males (69.4%). High rates of hypertension (86.1%), hyperlipidemia (84.4%), and coronary artery disease (35.6%) were observed. The most common presentation was stroke (48.4%), followed by TIA (31.8%) and amaurosis fugax (19.9%). The 30-day composite outcome of stroke, TIA, MI, or death was significantly lower in the early group (2.8%) compared to delayed (4.0%) and urgent (8.0%) groups (P=0.03). Thirty-day mortality was higher in the urgent group (2.9%) than in the early (0.3%) and delayed (0.3%) groups (P=0.01). On Cox regression, urgent timing (HR 2.50; P=0.044) and absent preoperative antithrombotic therapy (HR 3.75; P=0.006) were independent predictors of the 30-day composite outcome. At a median follow-up of 2.8 years, long-term MI was more frequent in the early group (4.7%) than in the urgent (1.5%) and delayed (1.6%) groups (P=0.03). Restenosis was highest in the urgent group (9.5%), followed by delayed (5.3%) and early groups (3.9%; P=0.05). Reintervention followed the same pattern (urgent 6.6% vs. delayed 3.1% vs. early 1.1%; P<0.01). On Cox regression, both urgent (HR 4.13; P=0.018) and delayed timing (HR 5.63; P=0.003) were independently associated with higher reintervention risk compared to early revascularization.

Conclusions. Early carotid revascularization have most favorable perioperative and long-term outcomes. Urgent revascularization carries significantly higher perioperative risk, while delayed intervention beyond 14 days reduces the potential preventive benefit of timely revascularization and is associated with higher long-term reintervention rates.

Keywords: Carotid artery stenosis, symptomatic carotid disease, carotid endarterectomy, carotid artery stenting, revascularization timing.

Cite as: Sanchez-Calderin D, Meschia JF, Kumar G, et al. Early Revascularization in Symptomatic Carotid Artery Disease Leads to Better Outcomes: A Ten-Year Multi-Center Cohort Analysis. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-18.

O-19 RAPID-FIRE Session II, talk 8 12:50 PM, 5 min Clinical Sciences

Impact of Extracorporeal Membrane Oxygenation as a Bridge to Lung Transplantation on Perioperative Outcomes, Renal Trajectory and Resource Utilization

Rijan Kafle, MBBS, Priyanka Poudel, Katherine Quiza, Suzit Bhusal, Ashlesha Chaudhary, Prasanth Balasubramanian, Remzi Bag, Brad A. White, Aashutosh Chaudhary, Pramod Guru, Sanjay Chaudhary

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Presenting author underlined. Principal investigator: Sanjay Chaudhary.

Presenter: Research Trainee, Department of Critical Care Medicine

Introduction. Extracorporeal membrane oxygenation (ECMO) is increasingly being used as a bridge to transplant (BTT) due to recent changes in lung allocation policy that prioritize critically ill candidates. We present a seven-year, single-center analysis of post-transplant outcomes comparing recipients bridged with ECMO to those transplanted without bridging support.

Methods. We performed a retrospective single-center study of adult lung transplant recipients from 2018–2024 using institutional EHR data. Patients were categorized by ECMO support at transplantation (Bridge ECMO vs No Bridge). The groups were compared in terms of baseline characteristics, operative factors, transfusion burden, and postoperative outcomes. Predictors of acute kidney injury (AKI) requiring dialysis for one year were found using multivariable logistic regression. Kaplan-Meier analysis was used to evaluate survival.

Results. Among 319 lung transplant recipients, 36 (11.3%) underwent transplantation while supported with ECMO. Bridged recipients were younger at transplant (58.6 vs 63.2 years, p=0.011) and required significantly higher perioperative transfusion (p<0.001). Postoperative complications were more frequent in the ECMO group, including acute kidney injury (72.2% vs 49.1%, p=0.009), infection/sepsis (27.8% vs 14.1%, p=0.034), and longer ICU stay (54.5 vs 10 days, p<0.001). Dialysis-requiring AKI at one year occurred in 50.0% vs 12.7% (p<0.001).

In multivariable analysis, post-transplant cardiogenic shock (aOR 7.49), serum creatinine at one month (aOR 11.27), and ICU length of stay (per 10 days, aOR 1.48; all p<0.001) independently predicted dialysis-requiring AKI, while ECMO bridging did not (aOR 2.74, p=0.281). One-year mortality (30.6% vs 29.3%, p=0.879) and post-transplant survival (HR 1.42, 95% CI 0.75–2.67) were similar.

Conclusions. ECMO bridging enabled transplantation accessible to selected groups of critically ill patients during this seven-year single-center experience. ECMO bridging was not associated with worse short- or intermediate-term survival, despite being linked to increased perioperative morbidity and resource utilization. This suggests that ECMO bridging is a viable treatment option but requires focused strategies to mitigate renal injury and perioperative morbidity.

Keywords: Extracorporeal Membrane Oxygenation (ECMO) Lung Transplantation ,Acute Kidney Injury, Renal Replacement Therapy, Critical Care Outcomes

Cite as: Kafle R, Poudel P, Quiza K, et al. Impact of Extracorporeal Membrane Oxygenation as a Bridge to Lung Transplantation on Perioperative Outcomes, Renal Trajectory and Resource Utilization. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-19.

O-20 RAPID-FIRE Session II, talk 9 12:55 PM, 5 min Clinical Sciences

Epilepsy as a Marker of Cognitive Decline in Cerebral Amyloid Angiopathy: Evidence from a Large Multi-Institutional Cohort

Ahmed Saad, MD, Yugant Khand, Sai Krishna Vallamchetla, Anshum Patel, Adrian Safa, Caroline P. Tatit, Gregory S. Day, Michelle P. Lin, Cady K. Block, William O. Tatum IV, Brin Freund

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Presenting author underlined. Principal investigator: Brin Freund.

Presenter: Research Trainee, Department of Neurology

Introduction. Cerebral amyloid angiopathy (CAA) is an amyloid mediated cortical and leptomeningeal small-vessel disease that commonly presents with lobar intracerebral hemorrhage, cognitive impairment, and transient neurological symptoms. Epilepsy occurs in 20-33% of patients with CAA, but their association with cognitive function and survival remains incompletely defined.

Methods. We conducted a retrospective cohort study using the Mayo Clinic Platform, which contains multi-institutional de-identified longitudinal electronic health records. Adult patients (≥18 years) with CAA were identified using ICD-10 code I68.0 and natural language processing of clinical notes including Boston criteria 2.0. Epilepsy was defined per ILAE criteria with ICD-10 codes and documentation in clinical notes, and cognitive dysfunction diagnosis was confirmed using patient history. Propensity score matching (1:1) was performed at the date of CAA diagnosis. Time-to-event outcomes were estimated using Kaplan-Meier methods and Cox proportional hazards regression for 60 months follow-up.

Results. Among the 2,717 patients with CAA (mean age 74.5 years), 733 patients (27.0%) had epilepsy. 673 propensity score–matched pairs were included in the final analysis. Epilepsy was associated with a higher risk of cognitive dysfunction (HR 1.34, 95% CI 1.13-1.60). No statistically significant difference was noted in all-cause mortality between cohorts (HR 1.07, 95% CI 0.86-1.33).

Conclusions. Epilepsy in patients with CAA was associated with higher risk of cognitive decline. There was no difference in all-cause mortality, but the confidence interval does not exclude a clinically meaningful effect in either direction. These findings highlight patients with epilepsy as a subgroup of CAA cases with increased risk for cognitive decline. Prospective studies with validated cognitive assessments are needed to confirm this association.

Keywords: cerebral amyloid angiopathy, seizures, epilepsy, cognitive decline, intracerebral hemorrhage

Cite as: Saad A, Khand Y, Vallamchetla SK, et al. Epilepsy as a Marker of Cognitive Decline in Cerebral Amyloid Angiopathy: Evidence from a Large Multi-Institutional Cohort. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-20.

O-21 FULL ORAL Session II, talk 10 1:00 PM, 10 min Clinical Sciences

Risk of Incident Seizures and Epilepsy After a Diagnosis of Idiopathic Intracranial Hypertension: A Matched Retrospective Cohort Study

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Presenting author underlined. Principal investigator: Brin Freund.

Presenter: Research Trainee, Department of Neurology

Background. Idiopathic intracranial hypertension (IIH) is primarily characterized by headache and visual dysfunction, yet accumulating evidence suggests broader neurologic manifestations, including skull base remodeling, temporal encephaloceles, and seizures. Whether IIH is associated with an increased risk of seizures or epilepsy remains uncertain. We evaluated the incidence of seizures or epilepsy after an IIH diagnosis compared with matched individuals without IIH and assessed whether any observed association could be explained by differential ascertainment.

Methods. We conducted a retrospective matched cohort study across all three Mayo Clinic sites. Adults with IIH were identified using diagnostic codes supported by lumbar puncture findings or repeated coding, with at least 12 months of pre-index records and follow-up. Patients with prior seizures or epilepsy were excluded. Comparators without IIH were matched on age, sex, and body mass index and were drawn from the same health system. Incident seizure or epilepsy was defined using validated coding criteria and antiseizure medication records. The primary outcome was the 3-year cause-specific hazard ratio, with death treated as a competing risk. A prespecified negative-control framework was used to evaluate potential surveillance bias.

Results. Primary cohort included 2,606 patients with IIH and 4,037 matched comparators (mean age 34.7 years, 87% female, mean BMI 36.3 kg/m^2). Seventy-four incident seizure or epilepsy events occurred during follow-up. Incidence rates were 9.38 per 1,000 person-years in the IIH group and 1.78 per 1,000 person-years among comparators. IIH was associated with a significantly increased risk of seizures or epilepsy (hazard ratio 5.58, 95% CI 3.15-9.87, $p<0.001$). Three-year cumulative incidence was 2.54% versus 0.49%, corresponding to an absolute risk increase of 2.05 percentage points. Negative-control analyses suggested that baseline imbalances and surveillance bias were unlikely to fully explain the observed association.

Conclusions. Patients with IIH exhibited a higher incidence of seizures and epilepsy than matched controls, with approximately a fivefold relative increase in risk. Although causality cannot be established, the findings indicate that seizure risk warrants consideration in the clinical management of IIH and support prospective studies incorporating detailed imaging, cerebrospinal fluid measures, and electroencephalographic confirmation of seizure events.

Keywords: Idiopathic Intracranial Hypertension, Seizures, Epilepsy, Intracranial Pressure, Cohort Study

Cite as: Khand Y, Tarabeah S, Saad A, Tatum W, Freund B. Risk of Incident Seizures and Epilepsy After a Diagnosis of Idiopathic Intracranial Hypertension: A Matched Retrospective Cohort Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-21.

O-22 RAPID-FIRE Session II, talk 11 1:10 PM, 5 min Clinical Sciences

The Predictive Value of SHAPE Test in Predicting Perioperative Major Adverse Cardiac Events: A Single-Center Retrospective Analysis

Alexis J. Magana, MD, Tachy V. Hernandez Romero, Zulma Sosa Carcamo, Ridwan Ahmad, Rafael A. Landeros Centeno, Juan C. Cardenas Rosales, Joan M. Irizarry Alvarado

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Presenting author underlined. Principal investigator: Joan M. Irizarry Alvarado.

Presenter: Research Trainee, Division of General Internal Medicine

Background. Postoperative major adverse cardiac events (MACE) remain a leading cause of perioperative morbidity and mortality. The assessment of cardiovascular risk prediction before elective surgery continues to be challenging for clinicians. The Submaximal Heart and Pulmonary Evaluation (SHAPE) Test is a noninvasive, cardiopulmonary stress test that measures exercise capacity and physiological response to exertion. It is used in the evaluation and management of patients with pulmonary hypertension. However, its role in predicting postoperative cardiovascular complications remains underexplored. This study evaluates whether SHAPE testing, regardless of indication, is associated with the incidence of postoperative MACE at 30, 90 days, and one year following surgery.

Methods. We retrospectively identified adult patients with ≥ 18 years of age who underwent elective procedures at a single healthcare system from January 1, 2021, to January 1, 2025. All patients completed a preoperative SHAPE Test within one year before surgery. Postoperative MACE were defined as myocardial infarction, heart failure, stroke, and death within 30, 90 days, and one year following surgery, as measures for the primary endpoint.

Results. A total of 88 patients (median age, 62; 68.2% women; median BMI, 29.6 kg/m2) were included. Postoperative MACE occurred in 12 (13.6%) patients within 1 year postoperatively. The median VO2max from the SHAPE test was 9.8 ml/kg/min (interquartile range, 9.0 – 13.5) among patients who experienced a MACE event, compared with 14.6 ml/kg/min (interquartile range, 12.8 – 16.7) among those who did not. We found an inverse relationship between VO2max and MACE, estimating an approximated 28% reduction in the odds of MACE for each 1 mL/kg/min increase in VO2max (OR 0.73, 95% CI 0.57 – 0.90, p=0.002).

Conclusions. Findings and the analysis suggest that SHAPE Test may improve preoperative clinical risk stratification before an elective noncardiac surgery. Further prospective studies in larger and more diverse populations are needed to support its application to improve surgical outcomes in preoperative care.

Keywords: preoperative assessment, postoperative cardiovascular events, submaximal cardiopulmonary exercise testing, Submaximal Heart and Pulmonary Evaluation (SHAPE) Test, postoperative cardiac risk.

Cite as: Magana AJ, Hernandez Romero TV, Sosa Carcamo Z, et al. The Predictive Value of SHAPE Test in Predicting Perioperative Major Adverse Cardiac Events: A Single-Center Retrospective Analysis. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract O-22.

POSTERS

Poster session I

10:30 AM to 11:30 AM. Abstracts for posters that accompany an oral presentation appear with the oral sessions.

P-01 POSTER Poster session I Cardiovascular, Transplant & Critical Care

Health-Related Quality-of-Life Measures in Patients with Heart Failure Cardiogenic Shock Following Axillary Mechanical Circulatory Support

Aarti Desai, Hans Mautong, Shriya Sharma, Jose Ruiz, Juan Leoni, Rohan Goswami

Division of Heart Failure and Transplantation, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Rohan Goswami.

Presenter: Aarti Desai, MBBS, Research Fellow, Division of Heart Failure and Transplantation

Background. Patients with end-stage heart failure-related cardiogenic shock (HF-CS) are conclusively associated with a poor health-related quality of life (HRQL). Axillary mechanical circulatory support (aMCS), such as the Impella 5.5, is increasingly used in this population and may improve HRQL during hospitalization by providing enhanced left ventricular unloading. We aimed to assess changes in HRQL between admission and two weeks after Impella 5.5 placement in patients with HF-CS, using the Kansas City Cardiomyopathy Questionnaire (KCCQ).

Methods. We conducted a prospective longitudinal analysis on patients with the Impella 5.5 between May 2023 and July 2023. Participants completed the condensed KCCQ-12 at admission and again two weeks post-implantation. Changes in the scores were evaluated using the Wilcoxon signed-rank test.

Results. Fifteen patients were enrolled. The median age was 59 years (50–63), and the median ejection fraction at implantation was 20% (15–30). On admission, most patients reported an overall HRQL of poor-to-fair (46.7%) according to the summary KCCQ-12 score. The median overall summary score increased significantly after Impella 5.5 support (50.52 vs. 28.13, $p = 0.005$). Symptom frequency (70.83 vs. 43.75, $p = 0.009$) and quality-of-life (50.00 vs. 12.50, $p = 0.023$) domains improved significantly, while physical limitation showed a positive trend and social limitation remained unchanged. These HRQL improvements occurred alongside a significant shift toward lower SCAI shock stages, marked increases in cardiac output and cardiac index, and no escalation in vasoactive-inotropic requirements.

Conclusions. Impella 5.5 support in HF-CS patients was associated with early and clinically meaningful improvements in HRQL, particularly in symptom frequency and quality of life, during the critical pre-transplant or recovery period. These findings suggest that the Impella 5.5 may provide both physiological and patient-perceived benefits in this high-risk population.

Keywords: quality of life, KCCQ, Impella, heart failure, transplantation

Cite as: Desai A, Mautong H, Sharma S, Ruiz J, Leoni J, Goswami R. Health-Related Quality-of-Life Measures in Patients with Heart Failure Cardiogenic Shock Following Axillary Mechanical Circulatory Support. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-01.

P-02 POSTER Poster session I Neuroscience & Neurodegeneration (basic)

Alpha-synuclein oligomers in Parkinson's disease with PRKN mutations, with and without Lewy-related pathology

Hiroaki Sekiya, Nathalie Goetz, **Noriko Akamine**, Asa Nakahara, Pamela J. McLean, Akiyoshi Kakita, Dennis W. Dickson

Department of Neuroscience, Mayo Clinic Florida, Jacksonville, Florida

Presenting author underlined.

Introduction. PRKN mutations are the most common cause of autosomal recessive early-onset Parkinson's disease (PD). Lewy-related pathology (LRP) is observed in only approximately one-third of patients with PRKN-associated PD (PRKN-PD), with no consistent common pathological substrate identified. αSYN oligomers, early aggregation species formed prior to Lewy body formation, are considered more toxic than mature fibrillar aggregates. However, conventional methods have limited sensitivity for detecting these oligomers, and whether they contribute to PRKN-PD remains unclear. We investigated the distribution and severity of αSYN oligomers and their relationship with LRP in patients with PRKN-PD.

Methods. We included seven patients with biallelic PRKN mutations, comprising point mutations, exonic deletions, and duplications/triplications in various combinations (three homozygous and four compound heterozygous). Clinical information, including age at onset and death, and disease duration, was collected. LRP was assessed by conventional immunohistochemistry using a phosphorylated αSYN antibody, while αSYN oligomers were examined by αSYN proximity ligation assay staining. Both pathologies were evaluated in the brainstem, limbic, and neocortical regions. The severity of αSYN oligomers was semi-quantitatively assessed based on the neuropil and neuronal staining.

Results. LRP was absent in four patients; among the remaining three, LRP extended to limbic structures in one, was limited to the brainstem in one, and was restricted to the dorsal motor nucleus of the vagus and locus coeruleus in one. In six patients, αSYN oligomer burden was higher in limbic and neocortical regions than in the brainstem; one patient, who carried the only compound heterozygous mutation combining a point mutation and an exonic deletion, showed minimal oligomer burden throughout. αSYN oligomer burden was comparable between patients with and without LRP. Disease duration showed a tendency toward a negative correlation with overall αSYN oligomer burden (r=-0.71; P=0.07).

Conclusion. αSYN oligomers were widely distributed across brain regions in PRKN-PD patients, regardless of LRP status. Although mitophagy impairment is considered the primary pathogenic mechanism in PRKN-PD, these findings indicate that αSYN aggregation, in the form of oligomers, also occurs in this disease and may represent a potential therapeutic target.

Cite as: Sekiya H, Goetz N, Akamine N, et al. Alpha-synuclein oligomers in Parkinson's disease with PRKN mutations, with and without Lewy-related pathology. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-02.

P-04 POSTER Poster session I Cardiovascular, Transplant & Critical Care

Redefining heart transplant eligibility – Heart-After-Liver Transplant (HALT) as an immunologic desensitization strategy in highly sensitized patients

Andrea Muniz, Jose Ruiz, Rohan Goswami, Laxmi Raj Bangari, Surbhi Dadwal, Aarti Desai, Piotr Wlodkowski, Andrew Keaveny, Mohamed Erefai, Jose Nativi-Nicolau, Basar Sarreyupoglu, Shennen Mao, Parag Patel

Division of Heart Failure and Transplantation, Mayo Clinic, Jacksonville, Florida

Presenting author underlined. Principal investigator: Rohan Goswami.

Presenter: Andrea Muniz, MD, Visiting Research Fellow, Cardiac Transplant

Background. Heart transplant candidates face limited donor availability and even more so in highly sensitized patients. This can be further complicated due to current desensitization strategies being frequently ineffective, expensive, and risky. The liver possesses unique immunologic properties that may facilitate antibody reduction in these high-risk individuals. We present a case series of heart-after-liver transplant (HALT) as an innovative desensitization strategy.

Methods. We conducted a single-center retrospective case series of consecutive adult patients with end-stage heart failure and high allosensitization who underwent HALT as a desensitization strategy.

Discussion. HALT may represent a feasible desensitization strategy for highly sensitized heart transplant candidates with otherwise limited donor access. The immunomodulatory properties of the liver allograft, combined with aggressive peri-operative desensitization and immunosuppressive therapy, may help overcome significant immunologic barriers and facilitate successful transplantation

Conclusions. Three highly sensitized heart transplant candidates (cPRA >99%) underwent successful transplantation with rapid DSA suppression and no AMR with moderate-term follow-up despite immunologic profiles that would traditionally be considered nearly prohibitive. Our experience demonstrates the potential for successful transplantation in patients traditionally considered prohibitively high risk. Larger multicenter studies are needed to validate these findings and define long-term outcomes.

Keywords: heart failure, heart-after-liver transplant, desensitization

Cite as: Muniz A, Ruiz J, Goswami R, et al. Redefining heart transplant eligibility – Heart-After-Liver Transplant (HALT) as an immunologic desensitization strategy in highly sensitized patients. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-04.

P-07 POSTER Poster session I Neuroscience & Neurodegeneration (basic)

Development and Validation of CRISPR Interference to Reduce Alpha-Synuclein Expression as a Therapeutic Strategy for Synucleinopathies

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Presenting author underlined. Principal investigator: Pam J. McLean.

Presenter: Ayman H. Faroqi, Department of Neuroscience

Introduction. Synucleinopathies are progressive neurodegenerative disorders characterized by the aberrant accumulation of alpha-synuclein (αSyn). Elevated expression of αSyn, encoded by the SNCA gene, is central to pathogenesis; therefore, its downregulation represents a promising disease-modifying strategy.

Methods. A CRISPR interference (CRISPRi) system was developed and validated to repress SNCA transcription and decrease αSyn levels. Following the design and in vitro screening of guide RNAs, a lead candidate was integrated into a viral vector for in vivo assessment in a humanized transgenic mouse model of synucleinopathy across neonatal and adult stages.

Results. Neonatal CRISPRi intervention achieved robust transcriptional repression, significantly reducing total, pathologically associated, and oligomeric αSyn. However, intervention in adult mice triggered an adaptive immune response in the mature central nervous system, highlighting a critical translational barrier.

Conclusions. This work provides proof-of-concept that CRISPRi-mediated suppression can recalibrate the protein landscape and prevent pathology in a humanized model. While overcoming immunological barriers in mature hosts remains a challenge, these findings establish a foundational framework for developing targeted, disease-modifying gene therapies for synucleinopathies.

Keywords: gene therapy, AAV, CRISPRi, synucleinopathies, alpha-synuclein

Cite as: Faroqi AH, Amerna D, Regalia D, et al. Development and Validation of CRISPR Interference to Reduce Alpha-Synuclein Expression as a Therapeutic Strategy for Synucleinopathies. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-07.

P-08 POSTER Poster session I Neuroscience & Neurodegeneration (basic)

Molecular and spatial dissection of orexinergic neuron subpopulation vulnerability in Alzheimer's disease

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Introduction. Sleep-wake disruption is an early feature of Alzheimer's disease (AD). Lateral hypothalamic orexinergic neurons (OrxN), which stabilize arousal, degenerate early in AD, but the mechanisms of selective OrxN vulnerability or persistence remain unclear.

Methods. We will apply Visium HD FFPE spatial transcriptomics to postmortem human hypothalamus spanning Braak stages 0, I, VI, and atypical AD. Serial sections labeled for orexin-A and CP13 will identify OrxN-enriched lateral hypothalamic regions, estimate tau burden, and align imaging features with Visium coordinates. Spots will be classified as OrxN-enriched, adjacent, or poor, followed by clustering and differential expression analyses to define OrxN states and associations with disease stage and tau.

Results. We previously showed substantial OrxN loss by Braak stage I, with ~40% neuronal loss early in disease and 20% of OrxN persisting in late-stage AD. Preliminary analyses in control and AD hypothalamic tissue identified differentially expressed markers among OrxN subpopulations, supporting the feasibility of resolving transcriptionally distinct OrxN states in human postmortem tissue. We expect these states to differ across AD progression and to reveal molecular signatures linked to early vulnerability or late resilience, including stress-response, metabolic, proteostatic, and signaling pathways.

Conclusion. This study will establish an initial spatial transcriptomic framework for understanding selective OrxN vulnerability in AD and will provide a foundation for future studies of how glial and metabolic environments contribute to OrxN degeneration or survival.

Cite as: Chen C, Satpati A, Pereira F, et al. Molecular and spatial dissection of orexinergic neuron subpopulation vulnerability in Alzheimer's disease. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-08.

P-09 POSTER Poster session I Health Services, Digital Health & Community

Supporting Symptom Recognition and Clinical Communication in Early-Onset Colorectal Cancer: Young Adults' Perceived Usability and Feasibility of an AI-Enabled mHealth Tool

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Introduction. Early-onset colorectal cancer (EOCRC) is rising among adults under 45, yet most remain ineligible for routine screening, making techquitable symptom recognition and timely evaluation tools essential for early detection. This study examined young adults' (YAs) perceptions of the usability and feasibility of a Smart Symptom Tracking (SST) Tool, an AI-powered, human-centered mHealth intervention designed to help YAs track gastrointestinal symptoms, access tailored education, and generate downloadable health reports for clinical visits.

Methods. YAs participated in think-aloud semi-structured interviews while navigating the high-fidelity SST Tool prototype. Participants (N=24) ranged from 19–45 years old (M=28.21, SD=7.27). Guided by the Health Information Technology Usability Evaluation Model (Health-ITUEM), we analyzed the transcripts using an AI-assisted deductive qualitative approach. Researchers first coded 20% of transcripts and used these data to train MAXQDA AI Assist. The full dataset was coded across four Health-ITUEM constructs, perceived a) user control, b) usefulness, c) impact, and d) ease of use, with research-team review at each step.

Results. Findings revealed strong alignment between participant perceptions and the four usability constructs. Participants described the tool as useful for guiding decision-making. Participants endorsed the SST Tool's value for agency, empowerment, symptom tracking and monitoring, communication support, tailored education, personalization, and downloadable health reports. Areas for refinement included information overload, privacy, trust in AI-generated insights, customization, integration with healthcare platforms, and barriers to adoption.

Conclusion. Findings provide early evidence that the SST Tool is a feasible and usable AI-enabled mHealth system for supporting EOCRC symptom recognition, communication, and clinical advocacy among YAs outside routine screening pathways. The SST Tool offers a promising symptom-driven pathway for earlier detection. Future work should prioritize full-system development, stronger personalization features, transparent AI safeguards, and randomized controlled trial pilot testing against standard care.

Cite as: Franco CL, Kissinger-Knox A, Kauer T, et al. Supporting Symptom Recognition and Clinical Communication in Early-Onset Colorectal Cancer: Young Adults' Perceived Usability and Feasibility of an AI-Enabled mHealth Tool. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-09.

P-10 POSTER Poster session I Surgery, Wound Care & Orthopedics

Complex Wound Healing Using APLICOR 3D Printed Fat Graft: An Early Institutional Case Series

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Presenter: Dhruva Kumar Lokesh, Department of Surgery

Background. Autologous fat grafting is a recent topic of interest in regenerative medicine, with its use being limited by processing speed and mode of delivery. APLICOR 3D system is an advancement in this field that processes autologous adipose tissue into customized grafts tailored to wound geometry.

Methods. We conducted a retrospective single center, single surgeon study between January 2025 and January 2026 of patients who underwent autologous fat grafting (AFG) using APLICOR 3D to create transplantable adipose graft. Image capture software was used to determine the required adipose volume to print a customized 3D graft. Adipose tissue was extracted using lipectomy, filtered with "adinizer" devices and washed with saline to isolate the 3% stromal vascular fraction (SVF) present in the supernatant, forming the "bioink." Grafts were printed with bioink utilizing a freezing method or intermixing with fibrin and thrombin to reinforce mechanical integrity, malleability for enhanced wound cohesion, ease of use and application. Grafts were directly applied to surgically prepared wound beds in the clinic or the operating room with an overlying nonadherent primary dressing followed by an absorptive secondary dressing.

Results. Ten patients of which seven were male and three were female, with thirteen wounds underwent AFG using APLICOR. Wounds ranged from extremity, sacrum, scalp, and thoracotomy wounds. Seven wounds were completely/almost healed (>90% healed) the rest have healed partially (50-90% healing), minimally healed (<50%) with only 1 patient not healing.

Conclusions. Bio-printed AFGs using APLICOR 3D supplemented with standard wound care shows promising results for complex chronic wounds.

Keywords: Fat grafting, 3D bioprinting, Chronic wounds, Mesenchymal Stem Cells, Wound healing

Cite as: Thiyagarajan SK, Garner S, Lokesh DK, Clements N, Price T. Complex Wound Healing Using APLICOR 3D Printed Fat Graft: An Early Institutional Case Series. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-10.

P-11 POSTER Poster session I Surgery, Wound Care & Orthopedics

Filling the Gap: A Novel Approach to Crohn's Perianal Fistula Closure Using APLICOR 3D Printed Stromal Vascular Fraction Graft

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Introduction. Perianal fistulas (including rectovaginal fistulas) are a severe manifestation of Crohn's disease with 17-50% of patients suffering from it. Two thirds of these patients require surgery with a third of them having recurrence. A method of using adipose derived mesenchymal stem cells in healing fistulas was described by Garcia-Olmo et al. We describe a novel technique of using the APLICOR 3D printed adipose derived mesenchymal stem cell rich stromal vascular fraction graft to plug and heal a chronic Crohn's rectovaginal fistula in a 23-year-old woman.

Methods. 20 ml of adipose tissue was harvested via standard lipectomy and passed through "adinizers" and washed with saline. The supernatant was allowed to settle to isolate the stromal vascular fraction (SVF). After internal closure of the fistula, 1 ml of the SVF was injected at every centimeter into the subcutaneous tissue around the tunnel. Then, 1.3 ml of the SVF was mixed with 0.4 ml of fibrin to print a fat graft of 5 sq.cm using the APLICOR 3D bioprinter. This was placed within the fistula, followed by affixing the graft by suturing the skin.

Results. At the two-week clinic visit there was complete closure of the fistula with no leakage reported by the patient post-operatively. This remained true following the two-month visit as well.

Conclusions. Autologous fat grafting has been shown to accelerate healing in chronic wounds due to their paracrine properties that enable angiogenesis, differentiation potential and immunomodulatory effect. This study strengthens the use of cutting-edge technology of adipose derived mesenchymal stem cells in Crohn's disease while advancing this concept with use of 3D printed SVF graft to plug and heal the fistula. Given the success of this treatment in a chronic rectovaginal fistula that was refractory to medical management, we aim to further refine this technique and conduct interventional clinical trials comparing outcomes with standard treatments.

Keywords: Mesenchymal Stem Cells, fat grafting, Crohn's disease, Perianal fistula, 3D bioprinting

Cite as: Lokesh DK, Deleon M, Stochhi L, Price T. Filling the Gap: A Novel Approach to Crohn's Perianal Fistula Closure Using APLICOR 3D Printed Stromal Vascular Fraction Graft. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-11.

P-13 POSTER Poster session I Women's Health, Weight & Shared Decisions

The Development and Pilot Testing of a Shared Decision-Making Encounter Aid for Weight Management Decisions

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Background. Obesity management involves multiple treatment options with differing benefits, risks, costs, and burdens, making these decisions highly preference sensitive. Shared decision-making (SDM) is recommended in obesity care, and yet practical tools to support real-time SDM during clinical encounters remain limited. We developed and pilot-tested a point-of-care decision aid, Weight Management Choice, to support structured discussions about weight management options.

Methods. This was a two-phase study combining a human-centered design process with a subsequent single-arm observational pilot study conducted in outpatient clinics at two Mayo Clinic sites. The design phase involved evidence review, observation of clinical encounters, iterative prototyping, and stakeholder-informed field testing. The pilot phase then evaluated the resulting tool in routine encounters between adults considering weight management options and participating clinicians. Post-encounter assessments included the SURE test, the 9-item Shared Decision Making Questionnaire (SDM-Q-9), and technology acceptance instrument assessing perceived usefulness (PU) and perceived ease of use (PEOU).

Results. Five clinicians were enrolled, four completed at least one assessment. Of 40 patients enrolled, 29 completed at least one patient-reported assessment (median age, 43 years; 75.9% female). Among patients completing the SURE test (n=27), the mean score was 3.81 (SD 0.77) on a 0–4 scale. Among those completing the SDM-Q-9 (n=29), the mean score was 75.0 (SD 18.2) on a 0–100 scale. Patients completing the technology acceptance instrument (n=26) reported mean PU and PEOU scores of 27.2 (SD 4.52) and 27.8 (SD 3.93), respectively. Across 33 clinician encounter-level assessments, corresponding mean scores were 27.5 (SD 2.74) and 29.1 (SD 2.74) on 5–35 scales.

Conclusions. The Weight Management Choice aid was perceived as feasible and had favorable immediate post-encounter ratings of shared decision-making, decisional conflict, and perceived usability. Future research should evaluate its impact on longer-term decision quality, treatment adherence, and clinical outcomes across diverse care settings.

Keywords: Weight Management, Obesity, Decision Aid, Endocrinology, Pilot Testing

Cite as: Abu Dabrh AM, Gala KS, Yu J, et al. The Development and Pilot Testing of a Shared Decision-Making Encounter Aid for Weight Management Decisions. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-13.

P-14 POSTER Poster session I Health Services, Digital Health & Community

Aligning Expectations in Community Advisory Board - Research Team Partnerships: Recruitment as an Opportunity for Early Partnership Building

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Background. Community advisory boards (CABs) strengthen community-engaged research by incorporating community perspectives. However, little is known about how recruitment framing shapes expectations before participation. This study compared CAB and research team perspectives to identify expectation mismatches and opportunities for early alignment.

Methods. We conducted a comparative qualitative analysis within a seven-year CAB–research team partnership focused on developing and reviewing a website providing cancer clinical trial information and research opportunities to diverse populations. Data included four semi-structured CAB member interviews, a focus group with five CAB members, and open-ended questionnaires completed by three research team members. Data was analyzed to identify areas of variation across participant perspectives.

Results. Four dimensions of expectation mismatch emerged: role, power, impact, and workflow. In defining their role, CAB members described themselves primarily as advocates for their communities and expected broader involvement across the research process, while the research team viewed the CAB's role more specifically as providing feedback on research endeavors. Differences in power centered on CAB members' expectations that they should provide meaningful influence over research decisions, while the research team viewed decision-making as constrained by funding, staffing, and regulatory requirements. Expectations around impact also differed, as CAB members emphasized immediate and visible community benefits, while the research team understood impact more broadly through longer-term sustainability and reach. Finally, differences in workflow reflected CAB members' expectations that recurring meetings would provide opportunities to understand how their contributions were incorporated into ongoing work, while the research team described CAB engagement within the constraints of competing research responsibilities.

Conclusions. Differences in expectations about roles and influence may arise even within sustained community-engaged research partnerships. Finding highlight recruitment as a potentially important and understudied opportunity for developing shared expectations. Three potential strategies emerged for strengthening this process: tailored recruitment messaging that clearly describes the CAB role, community-informed development of recruitment materials, and transparent communication about the scope of CAB participation.

Keywords: Community engaged research, Community advisory boards

Cite as: Moss F, Kauer T, Wang F, et al. Aligning Expectations in Community Advisory Board - Research Team Partnerships: Recruitment as an Opportunity for Early Partnership Building. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-14.

P-16 POSTER Poster session I Neuroscience & Neurodegeneration (basic)

Tau acetylated PHF6 reactive T cells presented by HLA-DRB1*04:04 in Alzheimer's Disease

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Introduction. Neurofibrillary tangles (NFTs) are hallmarks of Alzheimer's disease (AD) and formed by aggregation of the microtube-associated protein tau that undergoes many modifications, including acetylation on lysine (K) in tau. A piece of tau consisting of six amino acid residues, paired helical filament (PHF)6 (VQIVYK), plays a predominant role in tau aggregation. Merging evidence supports adaptive immune system plays an important role. However, the pieces of antigens, the types and functions of T cells, and HLA types associated with AD remain largely unknown.

Methods. We analyzed HLA associations in ~176,000 individuals with PD or AD versus controls across ancestry groups. We also compared postmortem brain density of NFTs and amyloid plaques in brain, tau and Aβ42 levels in cerebrospinal fluid (CSF) of ~8,000 individuals (controls and AD), and examined association of HLA in ~2,500 patient with pathologically demonstrated Lewy Body Dementia. This was followed by HLA binding, tetramer T cell studies and TCR activation by tau PHF6 in Jurkat cell.

Results. We found that HLA-DRB1*04:04 has protective effect in AD. We detected clonally expanded T cells recognizing tau, especially modified PHF6 with acetylation at lysine 311 (PHF6 aK311) when presented by DRB1*04:04, and the phenotypes of these T cells are distinct in AD patients versus cognitively healthy controls. Further, we identified different TCR clones from both AD patients and healthy controls only activated by tau PHF6 aK311 presented by DRB1*04:04.

Conclusion. An HLA-DRB1*04-mediated adaptive immune response against tau PHF6 aK311 decreases AD risk, offering the possibility of new therapeutic avenues.

Cite as: Luo G, Dodet P, Zhang J, et al. Tau acetylated PHF6 reactive T cells presented by HLA-DRB1*04:04 in Alzheimer's Disease. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-16.

Comparative Outcomes of En Bloc Resection, Definitive Radiotherapy, and Combined Modality Treatment for Sacral Chordomas: A Systematic Review and Meta-Analysis

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Introduction. Sacral chordomas are rare, locally aggressive notochordal tumors with an undefined optimal treatment strategy. En bloc resection with negative margins is the established first line approach, but local recurrence (LR) is common even after complete resection, and the role of definitive radiotherapy (RT) and combined modality remains unclear. We evaluated LR and metastatic outcomes in sacral chordomas across three treatment groups: surgery, definitive RT, and surgery with RT.

Methods. A systematic review and single-arm meta-analysis of proportions was performed per PRISMA guidelines (PROSPERO-registered). Studies of primary sacral chordoma were included if they reported extractable LR or metastatic outcomes for en bloc surgery with negative margins, definitive RT, or en bloc surgery plus RT. Pooled proportions were estimated using random-effects inverse-variance models with logit-transformed proportions. Long-term analyses included patients with 5 and 10-year follow-up

Results. Seventeen studies (479 patients and 150 LR events) were included in the analysis. Pooled LR proportion was 32.8% (95% CI, 23.3%–44.0%; $I^2 = 68.7\%$). By treatment subgroup, LR was 41.4% (95% CI, 26.8%–57.7%) for en bloc surgery, 24.8% (95% CI, 16.6%–35.4%) for definitive RT, and 17.6% (95% CI, 6.7%–39.0%) for en bloc surgery + RT, without a significant subgroup difference ($p = 0.096$). Metastatic disease was analyzed in 13 studies (364 patients, 78 events); the pooled metastasis proportion was 21.6% (95% CI, 15.4%–29.5%; $I^2 = 19.2\%$). By subgroup, metastasis was 22.3% (95% CI, 14.0%–33.6%) for en bloc surgery, 21.4% (95% CI, 11.7%–35.9%) for definitive RT, and 13.7% (95% CI, 2.8%–47.0%) for en bloc surgery + RT, without significant subgroup difference ($p = 0.813$). 5-year, pooled LR and metastasis proportions were 45.8% (95% CI, 27.4%–65.5%) and 28.5% (95% CI, 16.9%–43.9%) respectively, without significant differences. Ten-year, pooled LR and metastasis proportions were 62.7% (95% CI, 26.2%–88.9%) and 50.0% (95% CI, 14.8%–85.2%), respectively, without significant differences.

Conclusions. Despite en bloc surgery remaining the standard of care, definitive RT was associated with a numerically lower local recurrence rate than surgery alone (24.8% vs. 41.4%), with comparable metastasis rates between the two approaches. These differences did not reach statistical significance, and larger, prospective studies are needed to clarify the comparative efficacy between the treatment types.

Keywords: Sacral Chordoma, en bloc resection, definitive radiotherapy, local recurrence, metastases

Cite as: Iyer HG, Vega-Medina PA, Ghaderi S, Boktor B, Johnson WR IV, Akinduro OO. Comparative Outcomes of En Bloc Resection, Definitive Radiotherapy, and Combined Modality Treatment for Sacral Chordomas: A Systematic Review and Meta-Analysis. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-17.

P-22 POSTER Poster session I Cardiovascular, Transplant & Critical Care

Interventions for Anemia Correction in Critically Ill Lung Transplant Candidates

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Introduction. Anemia is a common finding among lung transplant candidates. However, whether chronic pre-transplant anemia independently predicts perioperative red blood cell (RBC) transfusion burden is unclear. This study aims to determine the impact of pre-transplant anemia severity and anemia-directed therapy on perioperative RBC transfusion burden in lung transplant recipients.

Methods. We conducted a retrospective cohort study of 404 adult lung transplant recipients. Anemia severity was defined by the lowest hemoglobin within 6 months before transplant and categorized as ≥12g/dl and 10-12 g/dL, 8–<10 g/dL, or <8 g/dL. Multivariable linear and logistic regression models adjusted for age, sex, body mass index, donor age, and pre-transplant creatinine.

Results. In our study, hemoglobin level was <10 g/dL in 58 patients (14.36%), including 14 (3.71%) with Hb <8 g/dL. Preoperative transfusions accounted for 30 cases (7.43%), intraoperative transfusions for 235 cases (58.17%), and postoperative transfusions for 267 cases (66.09%). Mean RBC transfusion was 0.04 ± 0.35 units preoperatively, 0.75 ± 1.81 units intraoperatively, and 2.22 ± 5.77 units postoperatively. Transfusion burden increased progressively with worsening anemia in unadjusted analyses. Iron therapy was used in 19 (4.70%) patients and erythropoiesis-stimulating agents in 5 (1.24%; 1 epoetin, 4 darbepoetin) with higher use among patients with severe anemia. Among patients with hemoglobin <10 g/dL and <8 g/dL, IV iron was associated with higher unadjusted transfusion, while ESA showed no association with PRBC use. In adjusted models, neither anemia severity nor anemia-directed therapy independently predicted transfusion burden. Pre-transplant creatinine was the only independent predictor (OR 2.00, 95% CI 1.27–3.07; p = 0.002).

Conclusion. In our cohort, RBC transfusion occurred mainly intraoperatively and postoperatively. Anemia-directed therapy was infrequently used and did not independently reduce transfusion, even in patients with hemoglobin <10 g/dL or <8 g/dL.

Cite as: Poudel P, Kafle R, Quiza K, et al. Interventions for Anemia Correction in Critically Ill Lung Transplant Candidates. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-22.

P-23 POSTER Poster session I Cancer & Translational Biology

The Small GTPase RAB27B Is Required for NSCLC Tumorigenesis and Promotes Cell-Autonomous Wnt Signaling in Cancer Stem Cells

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Introduction. Cancer stem cells (CSCs) contribute to non-small cell lung cancer (NSCLC) tumor initiation, progression, metastasis, therapeutic resistance, and recurrence and are sustained by aberrant developmental signaling. Our prior studies identified RAB27B, a small GTPase involved in vesicle trafficking and extracellular vesicle secretion, as upregulated in NSCLC CSCs and required for stem-like phenotypes. However, its role in lung tumorigenesis and cell-autonomous CSC signaling remains unclear.

Methods. RAB27B expression was evaluated in human lung adenocarcinoma and across lesion progression in the KrasG12D;Trp53fl/fl (KP) mouse model. Primary NSCLC datasets were analyzed to identify stem cell-associated pathways enriched in RAB27B-high tumors. RAB27B was knockdown in NSCLC CSCs to assess its role in CSC maintenance, transformed growth, invasion, and tumorigenicity. Rab27b was also deleted in alveolar type II (ATII) cells in a conditional KP mouse model to evaluate the requirement of Rab27b in lung tumorigenesis. Following identification of Wnt/β-catenin signaling as a RAB27B-associated pathway, β-catenin abundance and localization, TCF-dependent transcriptional activity, Wnt target gene expression, and WNT5A/FZD3 dependency were assessed upon RAB27B knockdown in NSCLC cells.

Results. RAB27B was elevated in human lung adenocarcinoma, associated with reduced patient survival, and increased during KP tumor progression. Pathway analysis of RAB27B-high NSCLC tumors identified enrichment of stem cell-associated pathways, including Wnt/β-catenin signaling. RAB27B depletion impaired CSC maintenance, clonal expansion, transformed growth, invasion, tumor growth, and metastatic activity. Rab27b deletion blocked KP-mediated transformation of ATII cells, reduced lung tumor burden, and improved survival. Mechanistically, RAB27B knockdown reduced β-catenin abundance and nuclear accumulation, TCF-dependent transcriptional activity, and MYC, CCND1, and LEF1 expression. Exogenous WNT3A restored β-catenin-dependent transcriptional activity in RAB27B-depleted cells. Knockdown of WNT5A or FZD3 reduced transformed growth, supporting a cell-autonomous role for Wnt signaling in CSCs.

Conclusion. RAB27B is required for oncogenic transformation, lung tumor progression, and NSCLC CSC activity. Our study also identifies a novel cell-autonomous RAB27B/Wnt signaling axis which may serve as a potential therapeutic vulnerability in NSCLC.

Cite as: Pandya P, Meneses K, Murray N, et al. The Small GTPase RAB27B Is Required for NSCLC Tumorigenesis and Promotes Cell-Autonomous Wnt Signaling in Cancer Stem Cells. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-23.

P-24 POSTER Poster session I Cardiovascular, Transplant & Critical Care

Development of HLA De Novo Donor-Specific Antibodies is Associated with Increased Risk of Rejection and Cardiac Allograft Vasculopathy Following Heart Transplantation

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Introduction. Development of HLA de novo Donor-Specific Antibodies (dnDSA) is associated with adverse allograft outcomes. The aim of our study was to assess the relationship between the development of HLA dnDSA and clinical outcomes in heart transplant recipients (HTR).

Methods. Retrospective analysis of 195 heart transplant recipients between 2020 and 2023 with 3-year follow-up period. HLA dnDSA were assessed using a single-antigen bead assay. The incidence of acute cellular rejection (ACR), antibody mediated rejection (AMR), and cardiac allograft vasculopathy (CAV) was evaluated during follow-up.

Results. HLA dnDSA were detected in 32/195 (16.4%) HTR during the first year post transplant. Detection of HLA dnDSA was associated with higher incidence of CAV (31.3% vs. 14.1%; OR, 2.77; 95% CI, 1.16–6.59; P = 0.03) and ACR (ACR 43.8% vs. 18.4%; OR, 3.45; 95% Cl, 1.54-7.70; P = 0.004). AMR occurred more frequently in dnDSA positive recipients (6.25% vs. 3.07%; OR, 2.11; 95% CI, 0.40–11.2), but this association was not statistically significant (P = 0.32).

Conclusion. Development of HLA dnDSA was associated with significantly increased risk of CAV and ACR in HTR. Our results support routine HLA dnDSA monitoring as a tool to identify HTR at increased risk of immune-mediated allograft injury and heart allograft dysfunction.

Cite as: Tripathee S, Shrestha N, Muñoz-Herrera C, et al. Development of HLA De Novo Donor-Specific Antibodies is Associated with Increased Risk of Rejection and Cardiac Allograft Vasculopathy Following Heart Transplantation. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-24.

P-25 POSTER Poster session I Surgery, Wound Care & Orthopedics

Accuracy of Computer-Assisted-Navigation in Total Ankle Arthroplasty Surgery Using Computed Tomography

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Presenter: Taylor Rakauskas, Department of Orthopedic Surgery

Introduction. Computer-assisted navigation (CAN) offers dynamic intra-operative guidance in total ankle arthroplasty (TAA). While laboratory data suggests superior precision, in vivo evidence is lacking. This study used weight-bearing computed tomography (WBCT) to evaluate CAN accuracy of post-operative TAA implant position relative to the plan. We hypothesized that absolute discrepancies would be <3 mm or degrees.

Methods. Thirteen consecutive patients (Nov 2024–Nov 2025) underwent TAA using CAN. Pre-operative CT-based 3D models were used for templating. Post-operative WBCT scans were used to measure final implant seating, and computerized measurements were used to identify implant placement discrepancy. We analyzed absolute discrepancies between planned vs. final post-operative positions.

Results. Tibial absolute discrepancies were within desired limits for varus/valgus (1.65 ± 1.43 degrees), slope (1.87 ± 1.34 degrees), and cut height (0.68 ± 0.20 mm), but not for axial rotation (3.97 ±192.64). For the talus, varus/valgus (2.93 ± 1.86 degrees) and cut height (0.66 ± 0.71 mm) absolute20discrepancies were within desired limits. The increased talar slope deviation (3.90 ± 2.89) in the final position may be due to non-navigated implant impaction. Reduced standard deviations (SD) across all measurements underscore that CAN has potential to reduce procedural variability despite the surgeon's initial learning curve.

Conclusions. While discrepancies persisted in tibial rotational alignment and talar positioning, the low variability across measurements suggests that CAN may improve surgical consistency. These short-term positioning accuracy findings support the utility of CAN as a potential tool for enhancing implant positioning reproducibility in TAA.

Keywords: navigation, ankle, arthroplasty, accuracy

Cite as: Rakauskas T, El Masry S, Hamad C, Boux De Casson F, Haupt E. Accuracy of Computer-Assisted-Navigation in Total Ankle Arthroplasty Surgery Using Computed Tomography. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-25.

P-29 POSTER Poster session I Clinical Neurology & Neurosurgery

Cerebral Microbleeds and Risk of Cognitive Impairment and Mortality After Traumatic Brain Injury: A Real-World Retrospective Cohort Study

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Presenter: Yugant Khand, MBBS, Research Trainee, Department of Neurology

Introduction. Cerebral microbleeds (CMBs) are a marker of traumatic and vascular microvascular injury visualized on susceptibility-weighted or gradient-recalled echo MRI. While CMBs are linked to cognitive decline in other neurological populations, their prognostic significance after traumatic brain injury (TBI) remains poorly defined. We used a large real-world data platform to test whether TBI patients with documented CMBs have higher risk of incident cognitive impairment and mortality than TBI patients without CMBs.

Methods. We performed a retrospective cohort study using de-identified structured and unstructured EHR data in Mayo Clinic Platform. Patients with a diagnosis of TBI were identified via ICD-10-CM codes. Natural language processing (NLP) of clinical and radiology notes was used to identify mentions of microhemorrhage, microbleeds, or cerebral microbleeds. Cohort A comprised TBI patients with NLP-identified CMB documentation (n=1,996); Cohort B comprised TBI patients without such documentation (n=54,031). The index event was first TBI diagnosis. Incident cognitive impairment and all-cause mortality were assessed using Kaplan-Meier analysis with log-rank testing, censored at 61 months.

Results. Cohort A patients were older (mean age 60.3 vs 55.6 years) and more frequently identified as White (91.8% vs 76.5%) than Cohort B, sex distribution was similar. CMB presence was associated with significantly higher mortality (60-month survival 86% vs 89%; log-rank=11.26, p=0.00079; HR 1.24, 95% CI 1.09–1.41). The association with cognitive impairment was substantially stronger: 60-month cognitive-impairment-free survival was 71.4% in Cohort A versus 91.8% in Cohort B (log-rank=944.37, p<0.005; HR 3.73, 95% CI 3.40–4.08), with divergence between curves apparent within the first month of follow-up.

Conclusions. In this large real-world cohort, TBI patients with CMBs had nearly a four-fold increased hazard of incident cognitive impairment and a modestly increased hazard of mortality compared with TBI patients without CMBs. These findings support CMBs as a clinically meaningful post-traumatic imaging biomarker and suggest a role for routine susceptibility-sensitive MRI screening and closer cognitive surveillance in TBI patients with CMBs. NLP-based phenotyping combined with ICD-10-CM diagnosis codes offers a scalable approach for identifying imaging biomarkers and longitudinal neurological outcomes in real-world datasets.

Keywords: Traumatic Brain Injury; Cerebral Microbleeds; Cognitive Impairment; Cox Proportional Hazards; Mayo Clinic Platform

Cite as: Khand Y, Afia M, Barrett K. Cerebral Microbleeds and Risk of Cognitive Impairment and Mortality After Traumatic Brain Injury: A Real-World Retrospective Cohort Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-29.

P-30 POSTER Poster session I Clinical Neurology & Neurosurgery

EEG-Derived Brain Age as a Biomarker of Neurobiological Aging: A Longitudinal Multi-Pipeline Validation Study

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Introduction. Resting electroencephalography (EEG) offers a practical approach to quantifying brain aging, but its predictive performance and physiological interpretation require systematic evaluation. We benchmarked complementary modelling approaches and examined whether EEG derived brain age reflects oscillatory and aperiodic signatures of ageing.

Methods. We analysed baseline recordings from 608 adults in the Dortmund Vital Study, aged 20 to 70 years (61.8% women). EEG from 64 channels, recorded with eyes open and closed, underwent automated preprocessing. Models included spectral ridge regression, Riemannian covariance regression, convolutional networks, self supervised representations, pretrained EEG models, and stacked ensembles. Performance was assessed using participant level 10-fold cross validation and bootstrap confidence intervals. Spectral parameterisation separated oscillatory activity from the aperiodic background, and brain age gaps were adjusted for chronological age.

Results. The final ensemble achieved an uncorrected mean absolute error (MAE) of 6.53 years (95% CI 6.14 to 6.93), R^2 of 0.68, and correlation of 0.83 with chronological age. This represented approximately 49% lower error than the mean age baseline (12.76 years). The interpretable Riemannian model combining both recording conditions achieved an MAE of 7.16 years, outperforming the convolutional network (7.64 years) and frozen pretrained model (11.10 years). Predictions showed high agreement between recording half in 150 participants, with an intraclass correlation of 0.957. Spectral model permutation testing supported an EEG association with age ($p = 0.002$). Older adults showed flatter aperiodic spectra and reduced oscillatory alpha activity. Central aperiodic exponent differences between younger and older groups reached a mean absolute Cohen's d of 0.91. A larger brain age gap was associated with a lower central aperiodic exponent after controlling for chronological age (partial $r = -0.343$). Age restricted analyses detected predictive signal among adults aged 20 to 40 years ($p = 0.005$), while prediction within ages 60 to 70 did not outperform chance.

Conclusions. These findings support resting EEG as a candidate measure of brain ageing, combining useful age prediction with interpretable spectral correlates. External validation and longitudinal studies are needed to establish whether the brain age gap captures clinically meaningful differences beyond chronological age.

Keywords: EEG-Based Brain Age Prediction, Brain Aging Biomarkers, Electroencephalography, Aperiodic Neural Activity

Cite as: Khand Y, Colella FE, Abdelhameed A, et al. EEG-Derived Brain Age as a Biomarker of Neurobiological Aging: A Longitudinal Multi-Pipeline Validation Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-30.

P-31 POSTER Poster session I Surgery, Hepatobiliary Surgery

Predicting Early Readmission After Pancreatic Resection: Implications for Postoperative Discharge Planning

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Presenting author underlined. Principal investigator: John A. Stauffer.

Presenter: Ema Portal, MD, Research Trainee, Department of Surgery

Background. Early readmission after pancreatic resection remains a significant challenge. We sought to identify predictors of 30-day surgical readmission and assess their consistency across pancreaticoduodenectomy (PD) and distal pancreatectomy (DP).

Methods. We conducted a single-center retrospective cohort study of pancreatic resections performed between 2018 and 2024. The primary outcome was unplanned 30-day surgical readmission; secondary outcomes included any readmission and index-admission reoperation. Multivariable logistic regression identified independent predictors, with Firth penalization applied to procedure-specific models.

Results. Among 636 patients, 127 (20.0%) experienced surgical readmission. In multivariable analysis, ASA class IV (adjusted odds ratio [aOR] 8.51, 95% CI 2.27–31.59), postoperative pancreatic fistula (aOR 5.59, 95% CI 2.72–11.53), delayed gastric emptying (aOR 6.24, 95% CI 3.32–11.77), and abdominal abscess (aOR 15.21, 95% CI 7.52–32.32) independently predicted surgical readmission. Pancreatic fistula, delayed gastric emptying, and abdominal abscess remained significant in both PD and DP subgroups, without significant interactions by procedure type. Any 30-day readmission occurred in 149 of 653 patients (22.8%). An exploratory complication-based risk stratification demonstrated increasing surgical readmission rates from 7.4% without complications to 90.9% with all three complications (p for trend <0.001).

Conclusions. Pancreas-specific complications and poor preoperative physiologic status are strongly associated with early surgical readmission, with consistent findings across PD and DP. Identifying these complications before discharge may help guide risk-adapted postoperative surveillance and discharge planning.

Keywords: Surgical Outcomes; Textbook Outcomes; Pancreatectomy

Cite as: Portal E, Verastegui A, Muniz A, et al. Predicting Early Readmission After Pancreatic Resection: Implications for Postoperative Discharge Planning. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-31.

POSTERS

Poster session II

1:15 PM to 2:15 PM. Abstracts for posters that accompany an oral presentation appear with the oral sessions.

P-32 POSTER Poster session II Surgery, Hepatobiliary Surgery

Influence of Surgical Approach on Textbook Outcomes after Pancreaticoduodenectomy and Distal Pancreatectomy

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Presenter: Hisham Sinan, MD, Research Trainee, Department of Surgery

Background. Traditional postoperative metrics inadequately capture the complexity of pancreatic surgery, prompting adoption of the composite Textbook Outcome (TBO) as a more comprehensive benchmark. We evaluated the impact of surgical approach on TBO after pancreaticoduodenectomy (PD) or distal pancreatectomy (DP).

Methods. We performed a single-center retrospective study of all PD-DP resections performed at Mayo Clinic Florida between 2010–2024, excluding total pancreatectomy, unclassified resections, and conversions. The primary outcome, TBO, defined as absence of clinically relevant pancreatic fistula, postpancreatectomy hemorrhage, major complications, 30-day readmission, and in-hospital mortality. Multivariable logistic regression identified predictors of TBO failure, stratified by procedure type. Results are reported as adjusted odds ratios (aOR) with 95% confidence intervals.

Results. Among 1,218 resections (718 PD; 500 DP), TBO was achieved in 67.0% of PD and 84.8% of DP. In PD, TBO failure (33.0%) did not differ significantly by surgical approach (p=0.103). Liver fibrosis or cirrhosis (aOR 3.96, 95% CI 1.20–13.10; p=0.024) and vascular resection (aOR 1.69, 95% CI 1.02–2.82; p=0.044) independently predicted TBO failure, whereas pancreatic or periampullary adenocarcinoma was associated with lower odds (aOR 0.60, 95% CI 0.41–0.88; p=0.009). In DP, TBO failure occurred in 15.2% and differed significantly by approach (30.4% open, 11.6% laparoscopic, 25.7% robotic; p<0.001). Laparoscopic DP was independently associated with lower odds of TBO failure compared with open resection (aOR 0.31, 95% CI 0.16–0.61; p<0.001). Age was not independently associated with TBO failure in either cohort.

Conclusion. Our findings highlight that surgical approach influences TBO in DP but not pancreaticoduodenectomy, with laparoscopic DP achieving the highest composite outcomes and age alone not limiting success.

Keywords: Surgical Outcomes; Textbook Outcomes; Pancreatectomy

Cite as: Sinan H, Muniz A, Verastegui A, et al. Influence of Surgical Approach on Textbook Outcomes after Pancreaticoduodenectomy and Distal Pancreatectomy. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-32.

P-33 POSTER Poster session II Cardiovascular, Transplant & Critical Care

Association of ACP at ICU Admission with Outcomes and Clinical Interventions

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Presenter: Aaliya Rahman, Department of Critical Care Medicine

Objective. To assess the difference in 30-day mortality between patients who completed advanced care planning documentation prior to needing care in the intensive care unit (ICU) Secondary objectives included evaluating the utilization of life saving interventions (dialysis and inotropes) and code status changes.

Methods. This was a retrospective cohort study using electronic health record (EHR) data for adult ICU admissions between 2019 and 2024. For patients with more than one encounter during this period meeting the inclusion criteria of having had part of their care receipt in the ICU, only the first encounter was included. Multivariable logistic regression was used to evaluate the association between ACP and 30-day mortality, adjusting for age, sex, marital status, race, ethnicity, DNR/DNI status, code status changes, Charlson Comorbidity Index, dialysis initiation, and inotrope use. Separate models evaluated the association between dialysis and inotrope use, respectively, controlling ACP documentation in addition to the above listed covariates. Analyses were performed using R.

Results. This study included 14,310 patient encounters, of which 3226 (22.5%) had documented ACP prior to ICU admission. Patients with ACP were more likely than those without ACP to receive dialysis (12.7% vs 8.8%, $p < .001$) and inotropes (44.2% vs 33.2%, $p < .001$) after ICU admission. Controlling for age, sex, marital status, race, ethnicity, and Charlson Comorbidity Index, pre-ICU ACP documentation remained independently associated with higher odds of receiving both dialysis (aOR 1.17, 95% CI 1.02–1.34, $p = .021$) and inotropes (aOR 1.37, 95% CI 1.25–1.49, $p < .001$).

Conclusions. ACP before ICU admission was not associated with 30-day mortality but was linked to a greater likelihood of receiving dialysis and inotropic support; an association not explained by the comorbidity burden alone. These findings challenge the common assumption that ACP is associated with less aggressive end-of-life care and highlight the need for further research into the mechanisms underlying this relationship.

Keywords: Advance care planning, Intensive care unit, Goals of care, End of life care, Life saving intervention

Cite as: Rahman A, Diaz Milian R, Borkar S, Menser T, Guru P. Association of ACP at ICU Admission with Outcomes and Clinical Interventions. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-33.

P-37 POSTER Poster session II Pain, Rehabilitation & Musculoskeletal

Lumbar Bone Stress Injuries in Pitchers: A Scoping Review

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Presenting author underlined.

Introduction. Lumbar spine conditions impose a meaningful burden in baseball players. Lumbar bone stress injuries are particularly relevant in pitchers because repetitive extension and rotation may contribute to their development. Evidence guiding management and return to play remains limited.

Methods. This scoping review followed JBI guidance for conduct and PRISMA-ScR for reporting. An experienced medical librarian conducted a comprehensive literature search based on a prespecified protocol. Of 59 records identified, 18 full-text articles were assessed for eligibility, and seven met the inclusion criteria: original studies involving baseball players at any competitive level and primarily reporting lumbar bone stress injuries. Pitcher-specific findings and data on management, rehabilitation, and return to play were extracted when available. Data from the included studies were charted and descriptively synthesized.

Results. Seven studies were included. Most examined adolescent baseball players, with fewer involving professional athletes; all were conducted in the United States or Japan. Study designs were predominantly retrospective or cross-sectional, and most samples combined playing positions, limiting pitcher-specific synthesis. Lumbar bone stress injury definitions included early-stage spondylolysis identified by pedicle hyperintensity on MRI, pars interarticularis stress reaction or fracture, spondylolysis, spondylolisthesis, and pedicle stress fracture. MRI was the most frequently reported imaging modality, and L5 was the most affected level. Management and return to play were sparsely reported; when described, treatment was predominantly nonoperative. One professional cohort found that nearly all athletes returned to play, but pitchers required a longer median time than position players (69 vs 34 days). One laterality study found lesions more frequently contralateral to the throwing side in pitchers (32 vs 16), and one case-control study found greater lumbar lordosis among athletes with spondylolysis.

Conclusion. Existing evidence characterizes lumbar bone stress injuries in baseball players, but pitcher-specific management and return-to-play data remain limited. Treatment and rehabilitation protocols were inconsistently reported and lacked standardization. Future studies should develop and evaluate pitcher-specific management and return-to-play criteria to guide clinicians and improve athlete outcomes.

Cite as: Pagan Rosado R, Monge Ferrin A, Prokop L. Lumbar Bone Stress Injuries in Pitchers: A Scoping Review. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-37.

P-38 POSTER Poster session II Neuro-oncology

Atorvastatin-Mediated Targeting of Lipid Metabolic Reprogramming in Glioblastoma

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Presenter: Anna Nulk, PhD, Department of Molecular Medicine

Introduction. Glioblastoma (GBM) is the most common malignant brain tumor in adults, affecting approximately 5 in every 100,000 individuals with median survival only reaching ~15 months. The standard-of-care regimen consists of maximal surgical resection, followed by radiation and chemotherapy using temozolomide (TMZ). Our lab has demonstrated that therapy-resistant stem cell-like tumor cells, termed treatment resistant persister cells (TRPCs), display a disproportionate ability to uptake, store, and utilize lipids compared to treatment susceptible cells (TSCs). Thus, adaptations in lipid metabolism provide a novel mechanism of chemoresistance in TRPCs that current therapies do not target. Here, we examine the use of atorvastatin (ATV) to disrupt this emergent lipid metabolic adaptation and driver of GBM recurrence.

Methods. Lipid dynamics and cell viability were assessed in the KR158 murine glioma model along with patient derived GBM cell lines (n=3) using Nanolive holotomagraphy (n=3-4/group). Cells were labeled with CellTrace Far Red to differentiate TSCs and TRPCs. Hourly imaging was performed following treatment with ATV for 24 hours. Cell growth was assessed using CellTiter-Glo (n=4/group) at 24, 48, and 72 hours. C57BL/6 mice were implanted with 50,000 luciferase-expressing KR158 cells. Mice were treated with vehicle or ATV (n=10/group) for 3 cycles of 5 days on, 2 days off.

Results. In the L0 and L2 patient-derived cell lines, TRPC survival and lipid abundance were significantly diminished in the context of ATV. This was observed to a limited extent in KR158 and L1 cells. However, KR158 cell growth was significantly decreased with ATV treatment using CellTiter-Glo at all time points examined. Preliminary in vivo experiments with KR158 cells in C57BL6 mice revealed a significant survival benefit in ATV-treated animals compared to control.

Conclusions. ATV prompted a decrease in KR158 tumor cell growth in vitro along with a survival advantage in vivo. Heterogeneity in the extent of lipid metabolic remodeling was observed between the three patient-derived cell lines, suggesting differences in the reliance on lipid metabolism and potential response to statin therapy. Future experiments will explore ATV and TMZ combination treatment with models of GBM established with L0 and L2 patient-derived cells.

Keywords: Glioblastoma; Lipid Metabolism; Drug Resistance, Neoplasm; Hydroxymethylglutaryl-CoA Reductase Inhibitors

Cite as: Nulk AR, Redmon E, Searl N, et al. Atorvastatin-Mediated Targeting of Lipid Metabolic Reprogramming in Glioblastoma. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-38.

P-39 POSTER Poster session II Neuro-oncology

Hypofractionated Radiotherapy in Glioblastoma: A Review of Comparative Effects on Safety, Tumor Control, and Survival

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Presenter: Anvi K. Reddy, MBBS, Department of Radiation Oncology

Introduction. Glioblastoma (GBM) is the most common malignant primary central nervous system tumor and remains associated with poor outcomes despite advances in surgery, radiotherapy, and systemic therapy. Hypofractionated radiotherapy (HFRT) has emerged as an alternative strategy to shorten treatment duration while maintaining efficacy. More recently, dose-escalated and ultra-hypofractionated regimens have been investigated, raising important questions regarding optimal patient selection and the future role of treatment acceleration. We reviewed the existing literature to evaluate current evidence supporting HFRT, identify limitations of available studies, and assess the rationale for ongoing and future clinical trials.

Methods. Relevant English-language publications evaluating HFRT in newly diagnosed GBM were reviewed.

Results. Randomized studies consistently demonstrate that HFRT provides survival outcomes comparable to CFRT in elderly and poor-performance-status patients while substantially reducing treatment burden. Preclinical and clinical evidence suggests that hypofractionation may enhance antitumor immune responses through increased antigen release and immune modulation, supporting investigation of HFRT in combination with immune checkpoint inhibitors, vaccine therapies, targeted agents, and tumor-treating fields. However, evidence supporting HFRT in younger and fitter patients remains limited. The phase III GOLD trial failed to demonstrate non-inferiority of ultra-hypofractionated chemoradiation, reporting inferior overall survival and increased treatment-related toxicity compared with standard therapy. In contrast, ongoing trials such as SAGA and COmpACT are evaluating modern HFRT approaches incorporating advanced treatment planning, dose escalation, and broader patient inclusion criteria. Future research is expected to focus on biomarker-driven patient selection, advanced imaging-guided target delineation, radiobiologically optimized schedules, and integration with novel systemic therapies.

Conclusions. HFRT is an established treatment option for selected elderly and frail patients with GBM, providing comparable outcomes while reducing treatment burden. Ongoing randomized studies will determine whether accelerated and dose-escalated HFRT approaches can safely expand to broader patient populations while maintaining efficacy, minimizing toxicity, and improving patient-centered outcomes.

Keywords: GBM, glioma, sbrt, radiotherapy, review

Cite as: Reddy AK, Trifiletti DM, Lehrer E, et al. Hypofractionated Radiotherapy in Glioblastoma: A Review of Comparative Effects on Safety, Tumor Control, and Survival. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-39.

P-41 POSTER Poster session II Cancer & Translational Biology

Cutaneous Manifestations Secondary to CAR-T Cell Therapy: A Retrospective Cohort Study

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Introduction. Chimeric antigen receptor T-cell (CAR-T) therapy is one of the most innovative and rapidly emerging immunotherapies for the treatment of various types of blood cell malignancies. The cutaneous reactions and their concurrent histopathology associated with CAR-T therapy have not been extensively studied.

Methods. We conducted a retrospective chart review of all the patients who have received CAR-T therapy from June 12th, 2018, to May 18th, 2026, since the initiation of Mayo Clinic, Florida, CAR-T program database. We analyzed individual records for the presence of any cutaneous symptoms post infusion.

Results. Out of 303 patients who received CAR-T during this time, 9 were included in the final analysis based on follow-up documentation by the treating oncologist or dermatologist. Rash onset was classified as early (<30 days), intermediate (30-180 days), or late (>180 days). The median time to rash onset was 22 days. Among patients with early-onset eruptions, 2 patients (2/6, 33.33%) developed a blistering eruption, 2 patients (2/6, 33.33%) developed morbilliform eruption, one patient developed a morbilliform rash with one bullous lesion (1/6, 16.67%), and one patient developed abnormal ecchymosis (1/6, 16.67%). Only one of these patients' blistering rashes was diagnosed as Toxic Epidermal Necrolysis (TEN). Among patients with intermediate-onset eruptions, the erythematous morbilliform rash was most common (2/3, 66.67%).

Conclusion. Overall, this study suggests that skin findings secondary to CAR-T cell therapy are a potentially common but an under reported phenomenon, and referral to dermatology is crucial for getting accurate treatment and diagnoses for these reactions.

Cite as: Velusamy B, Josephson S, Alhaj Moustafa M, et al. Cutaneous Manifestations Secondary to CAR-T Cell Therapy: A Retrospective Cohort Study. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-41.

P-42 POSTER Poster session II Surgery, Wound Care & Orthopedics

Prevalence of Micronutrient Deficiencies in Patients with Chronic Wounds: A Descriptive Analysis of 344 Patients

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Presenting author underlined. Principal investigator: Dr. Thea Price.

Presenter: Dhruva Kumar Lokesh, Department of Surgery

Introduction. The role of nutrition in wound healing has been the subject of extensive research. Patients with chronic wounds are especially vulnerable to nutrient and micronutrient deficiencies due to the increased nutritional demands of their nonhealing tissues. Despite the increasing literature focusing on specific nutrients and their effects on wound healing, there is a notable lack of large-scale studies that assess the prevalence of micronutrient deficiencies across a wide-ranging chronic wound population.

Methods. We conducted a retrospective review on patients evaluated for chronic wounds with varied etiologies such as diabetic foot ulcers, pressure ulcers, post-operative wounds etc. from August 2024 to May 2026 at Mayo Clinic Florida. Micronutrient panels were obtained for zinc, magnesium, iron, ferritin, copper, folate, ceruloplasmin, ascorbic acid, vitamin D, selenium, vitamin A, vitamin B12, vitamin K, vitamin E, and pyridoxal-5-phosphate (vitamin B6) under the guidance of a physician nutritionist. Mean values and the proportion of patients meeting laboratory criteria for deficiency were calculated.

Results. Our cohorts median age was 64 years old (IQR 56,76), with 53% male (n=181) and 47% female patients (n=162), overall, predominantly Caucasian (89%). Magnesium deficiency was the most prevalent finding, present in 133 of 344 patients (38.7%), followed by iron deficiency in 104 patients (30.2%). Vitamin D deficiency was identified in 49 patients (14.2%) and zinc deficiency in 39 patients (11.3%). Ascorbic acid and selenium deficiencies were each present in approximately 8% of the cohort (27/344, 7.8%, and 26/344, 7.6%, respectively). Vitamin B12 deficiency was identified in 15 patients (4.4%). Deficiencies in ferritin, copper, folate, ceruloplasmin, vitamin A, vitamin K, vitamin E, and vitamin B6 were each identified in 7 or fewer patients (≤2.0%).

Discussion. In our study, magnesium and iron deficiencies were found to be highly prevalent, and a significant number of patients had deficiencies in vitamin D, zinc, vitamin C, and selenium; micronutrients with established roles in wound healing. These findings suggest that micronutrient deficiencies are common in this population and support consideration of routine micronutrient screening as part of chronic wound evaluation with repletion and ongoing surveillance. Further research is needed to determine whether identifying and correcting these deficiencies improves healing outcomes.

Keywords: Nutrition, Chronic wounds, Diabetic foot ulcer, Micronutrient

Cite as: Lokesh DK, Muniz A, Price T. Prevalence of Micronutrient Deficiencies in Patients with Chronic Wounds: A Descriptive Analysis of 344 Patients. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-42.

P-43 POSTER Poster session II Surgery, Wound Care & Orthopedics

The effectiveness of Minimally Invasive Therapy for Hidradenitis Suppurative (MITHS) as a maintenance therapy in Hidradenitis Suppurativa: A novel treatment approach

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Presenter: Dhruva Kumar Lokesh, Department of Surgery

Introduction. Hidradenitis Suppurativa (HS) is a chronic, inflammatory skin condition affecting nearly 81 million people that is notorious for significant pain, tunneling, scarring, malodorous discharge adversely affecting the patients' work, social and sexual life. Due to its recurrence and refractory nature to medical treatment, radical excisions that leave large skin defects are the final resort. We present our experience with using a hydro-surgical minimally invasive treatment as a maintenance therapy in this condition.

Methods. MITHS involves debridement of the tunnels with VersajetTM hydro-surgical tool supplemented with laser ablation, botox and steroid injections as needed. We retrospectively analyzed the effectiveness of this therapy in all patients that received it in the time frame ranging from 18 months ago to 1 month ago. The subjective effectiveness was recorded via a verbal questionnaire answered by the patients on a phone call.

Results. A total of 51 patients received the minimally invasive therapy of which 33 (4 with Hurley's Stage II and 29 with Hurley's Stage III) patients answered the questionnaire. The average age of the cohort was 37.96 with majority having symptoms starting in the first two decades of life, the average number of years between symptom onset and diagnosis was 6.12 years. The average pain score before the procedure was 6.21 which was reduced to 3.13 after the procedure, with patients reporting an average of 73% resolution of symptoms and average remission period of 74 days.

Conclusions. Surgical excision which is the gold standard for management of HS often results in massive open wounds requiring weeks of extensive wound care, necessitating admission to wound care facilities or deployment of home health services and could still recur. This treatment protocol supplemented with medical management will help downstage Hidradenitis Suppurativa, resulting in longer remission periods, reduced frequency and severity of symptoms, improved quality of life, preventing radical excision and need for reconstructive surgeries. We found this technique to be non-inferior compared to deroofing as it excises the tunnels, followed by scarring leading to decrease in overlying sebaceous gland and hair follicle growth.

Keywords: Hidradenitis Suppurativa, minimally invasive therapy, Chronic disease, Outcomes research, Quality of Life

Cite as: Lokesh DK, Muniz A, Price T. The effectiveness of Minimally Invasive Therapy for Hidradenitis Suppurative (MITHS) as a maintenance therapy in Hidradenitis Suppurativa: A novel treatment approach. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-43.

P-48 POSTER Poster session II Cancer & Translational Biology

Pancreatic Findings on Prior Low-Dose Chest CT Examinations Before Pancreatic Ductal Adenocarcinoma Diagnosis: A Blinded Review Using RADPEER Criteria

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Presenting author underlined. Principal investigator: M. B. Wallace.

Presenter: Felix Lopez Dominici, MD, Department of Gastroenterology and Hepatology

Background. Pancreatic ductal adenocarcinoma (PDAC) is frequently diagnosed at an advanced stage despite prior imaging examinations. We evaluated whether blinded review of prior low-dose computed tomography (LDCT) examinations identified pancreatic abnormalities more frequently in patients who subsequently developed PDAC than in matched controls.

Methods. We performed a retrospective descriptive study including 13 patients who subsequently developed PDAC and 52 comparison patients selected for similar age and sex. A board-certified radiologist, blinded to clinical outcomes, independently reviewed all prior low-dose chest computed tomography (LDCT) examinations using the RADPEER scoring system while documenting predefined pancreatic imaging findings. Continuous variables were compared using Student's t-test or the Mann-Whitney U test, as appropriate, and categorical variables using Fisher's exact test.

Results. Sixty-five patients (13 PDAC cases and 52 comparison patients) were included. Cases and comparison patients were similar in age and sex. The median interval from prior LDCT to PDAC diagnosis was 10 months (IQR, 6–26). Most PDACs arose in the pancreatic head (46%), and 70% of staged patients had stage IV disease at diagnosis. Blinded review of all 65 examinations classified the pancreas as normal in 69%, insufficiently visualized in 5%, and abnormal in 26%. Among evaluable examinations, RADPEER discrepancies (score ≥2) were more frequent in PDAC cases than the comparison group (50% vs. 2%, $p = 0.0001$). Abnormal pancreatic findings were also more common in cases (58% vs. 20%, $p = 0.013$), including pancreatic masses (25% vs. 0%, $p = 0.006$) and common bile duct dilation (17% vs. 0%, $p = 0.035$).

Conclusions. Blinded review of prior LDCT examinations using RADPEER criteria identified significantly more interpretive discrepancies and pancreatic abnormalities in patients who subsequently developed PDAC than in the comparison group. These findings suggest that pancreatic abnormalities may be retrospectively identifiable on chest LDCT before PDAC diagnosis.

Cite as: Lopez Dominici F, Osagiede O, LeGout J, Bolan C, Wallace MB. Pancreatic Findings on Prior Low-Dose Chest CT Examinations Before Pancreatic Ductal Adenocarcinoma Diagnosis: A Blinded Review Using RADPEER Criteria. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-48.

P-49 POSTER Poster session II Pain, Rehabilitation & Musculoskeletal

Early Adoption of Suzetrigine After FDA Approval: Inpatient and Outpatient Prescribing Patterns at a Large Academic Center

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Presented by Shrekshya Khadka on behalf of the authors. Principal investigator: Christine Hunt.

Presenter: Shrekshya Khadka, Research Trainee, Department of Pain Medicine

Background. Suzetrigine is a first-in-class, non-opioid oral analgesic that selectively blocks NaV1.8 voltage-gated sodium channels and was approved by the Food and Drug Administration (FDA) in January 2025 for moderate-to-severe acute pain in adults. Early prescribing patterns remain poorly described.

Purpose. To evaluate early inpatient and outpatient suzetrigine prescribing patterns at a large academic medical center.

Methods. We performed a retrospective chart review of all patients prescribed suzetrigine at Mayo Clinic from April-December 2025. Demographic and clinical characteristics, prescribing department, dosing regimen, multimodal analgesia (MMA) co-prescribing, refill patterns, and adverse events were analyzed separately for inpatient and outpatient groups. MMA was defined as suzetrigine prescribed with at least 2 additional analgesics of different mechanisms within the same encounter. Continuous variables were compared using Mann-Whitney U tests and categorical variables using Pearson chi-square tests; $p<0.05$ was significant.

Findings. A total of 1,280 prescriptions were reviewed (821 inpatient, 458 outpatient). Multimodal analgesia co-prescribing was more common inpatient, including opioids, acetaminophen, gabapentin, ketamine, and local anesthetics (all $p<0.001$), whereas NSAID use was slightly higher in outpatient population ($p=0.048$). Colorectal surgery was the most frequent inpatient prescriber, while otorhinolaryngology led outpatient use. FDA-labeled loading doses were more common inpatient, while outpatient prescriptions often omitted loading. Indications differed by setting (inpatient: predominantly acute surgical pain; outpatient: mixed acute surgical and chronic pain; $p<0.001$). Refills were rare inpatient, but occurred regularly in outpatient. Adverse events were uncommon (2.9%) and did not differ by care setting.

Conclusions. Early suzetrigine prescribing differed between inpatient and outpatient settings in specialty use, MMA co-prescribing, dosing, indication, and refill patterns. These data provide an early benchmark for clinical integration.

Keywords: Acute pain, Multimodal Analgesia, sodium channel blockers, non-opioid medication, postsurgical pain

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P-51 POSTER Poster session II Clinical Neurology & Neurosurgery

Comparative Outcomes of Endonasal and Transoral Odontoidectomy: An Institutional Experience and Meta-Analysis

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Introduction. Irreducible ventral compression of the craniovertebral junction typically requires anterior decompression via odontoidectomy often with posterior stabilization. The transoral approach (TOA) has long been the standard, but the endoscopic endonasal approach (EEA) has more recently emerged as a less invasive alternative. We combined an institutional cohort with a systematic review and meta-analysis comparing perioperative and neurological outcomes between approaches.

Methods. We performed a retrospective cohort study of odontoidectomy with posterior stabilization at the 3 Mayo Clinic sites, combined with a systematic review and meta-analysis comparing baseline, operative, and postoperative variables. Pooled proportions were analyzed by approach using logit-transformed data and generalized linear mixed models, with subgroup comparisons and direct pairwise meta-analysis.

Results. The institutional cohort comprised 36 patients (10 EEA, 26 TOA). EEA patients were younger (p=0.023) and had higher rates of diabetes (p=0.039) and sleep apnea (p=0.010), while preoperative sensory deficit was more common in the transoral group (p=0.011). Complete odontoidectomy was achieved in 32 patients. Patient undergoing EEA had more posterior levels fused and had greater blood loss.

The meta-analysis included 23 unique studies comprising 549 patients including our cohort. Pooled neurological improvement was 95.5% for EEA and 91.1% for TOA, and adequate decompression was 99.2% and 98.4%, respectively. CSF leak was more frequent after EEA (6.3% vs 1.5%), while dysphagia was more frequent after TOA (12.0% vs 0.8%). Heterogeneity was low-to-moderate for neurological improvement (I^2=29.9%) and moderate for transoral dysphagia (I^2=48.4%) in the TOA group. Subgroup differences were non-significant for all outcomes, though CSF leak approached significance (Q=3.46, p=0.063). Pairwise meta-analysis, feasible only for neurological improvement and CSF leak, suggested lower odds of neurological improvement with endonasal surgery (OR 0.57, 95% CI 0.42–0.78) based on only two informative studies, and an imprecise CSF leak estimate (OR 2.64, 95% CI 0.02–328)

Conclusions. EEA and TOA odontoidectomy showed similar rates of decompression and neurological improvement, with no significant differences between approaches. The observed differences were mainly associated with corridor-specific morbidity. These findings may reflect a role for anatomy and comorbidity in corridor selection.

Keywords: Odontoidectomy, Endoscopic Endonasal approach, Transoral approach, basillar invagination

Cite as: Iyer HG, Vega-Medina PA, Ghaderi S, et al. Comparative Outcomes of Endonasal and Transoral Odontoidectomy: An Institutional Experience and Meta-Analysis. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-51.

P-53 POSTER Poster session II Health Services, Digital Health & Community

Economic Evaluation of Health and Wellness Coaching: A Systematic Review

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Presenter: Jiakun Yu, Research Trainee

Introduction. Health and wellness coaching (HWC) is increasingly used to support behavior change, chronic disease self-management and prevention. As healthcare systems emphasize value-based care and sustainable approaches to improving population health, understanding the economic value of HWC is increasingly important. Although clinical and behavioral outcomes of HWC have been studied, evidence regarding its economic impact remains poorly characterized. This review addresses this gap by synthesizing published evidence on the economic outcomes of HWC interventions.

Methods. We conducted a systematic review following PRISMA guidance. PubMed, Embase, ERIC, Scopus, PsycINFO and Web of Science were searched for English-language studies published January 2000-May 2026. Eligible interventions met all four core constructs of the NBHWC standards. Studies with follow-up ≥12 weeks and primary cost data were included. Outcomes included delivery cost, cost offsets, cost-effectiveness, cost-utility, cost-benefit/ROI and indirect costs.

Results. 848 records were identified and screened, 198 full texts assessed, 21 studies, representing 22 cohorts, met the inclusion criteria. Populations were heterogeneous, spanning COPD, HIV, obesity, smoking cessation, frailty and mobility limitation, serious mental illness, irritable bowel syndrome, post-arthroplasty rehabilitation and hospital-to-home transitions. Cost-related outcomes included intervention-delivery resources and costs, healthcare utilization, direct medical and social-care costs, health utilities and QALYs. Cost-effective analyses were conducted in 12 studies, 8 of which conducted cost-utility analysis using incremental cost-effectiveness ratios. Cost-effectiveness outcomes were expressed as incremental cost-effectiveness ratios, average cost-effectiveness ratios, cost per behavioral outcome, net monetary benefit, or probability of cost-effectiveness.

Conclusion. Among studies meeting standardized HWC criteria, economic outcomes varied substantially. The available evidence suggests that HWC may have potential to reduce healthcare costs and provide economic value in selected populations, but the heterogeneity of interventions, populations, economic outcomes, and reporting methods limits definitive conclusions regarding its overall cost-effectiveness or return on investment. Future research should prioritize standardized, high-quality economic evaluations to clarify the long-term value and scalability of HWC.

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P-55 POSTER Poster session II Health Services, Digital Health & Community

Precision Medicine Through the Understand Phase: Leveraging the UTS Framework to Design Personalized AI Orchestration in Inpatient Fall Prevention

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Introduction. Inpatient falls remain a major patient safety challenge despite prevention strategies. Multifactorial, patient-centered interventions are more effective than many single-component approaches, yet translating risk assessment into timely, individualized prevention remains difficult. Ambient artificial intelligence (AI) enables continuous monitoring, behavioral interpretation, and automated intervention. Frameworks for precision AI deployment remain underdeveloped; this abstract examines how the UTS Understand phase can guide patient selection and implementation of ambient AI for inpatient fall prevention.

Methods. We conducted a case study of ambient computer-vision AI implementation for fall prevention at a tertiary academic medical center. Using the Understand phase of the UTS framework, we evaluated technology capabilities, clinical workflows, electronic health record (EHR)-based eligibility criteria, and data requirements before deployment. Retrospective analysis examined eligibility signals, including Hester Davis (HD) fall-risk scores and physician activity restriction orders, in identifying patients who experienced inpatient falls. Workflow mapping and technology assessment identified implementation challenges and opportunities for precision intervention design.

Results. The ambient AI platform provided continuous room monitoring, behavioral detection, automated verbal redirection, silent monitoring, and configurable suppression logic. Retrospective analyses showed that an HD score ≥15 identified 69.5% of patients who experienced falls, highlighting limitations of documentation-based eligibility models. Expanded eligibility strategies incorporating activity restriction orders or prolonged risk windows increased fall-event coverage but also increased intervention exposure. Workflow evaluation identified sources of precision loss, including documentation lag, score volatility, inter-rater variability, and monitoring gaps during suppression periods. These findings revealed a mismatch between evolving patient risk and episodic clinical documentation. Future implementation models included delirium-adaptive eligibility, medication-responsive monitoring, behavioral signal-driven activation, and trajectory-based risk prediction.

Conclusion. Future fall prevention strategies should incorporate dynamic behavioral, clinical, and contextual signals rather than static documentation thresholds.

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P-57 POSTER Poster session II Neuroscience & Neurodegeneration (basic)

High Resolution In Situ Visualization of Astrocyte-Associated Nanoscale Structures Through Combined TurboID Labeling and Expansion Microscopy

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Presenter: Shaina Matveev, Department of Neuroscience

Introduction. Astrocyte-mediated intercellular communication, including interactions at tripartite synapses and extracellular vesicle (EV)-mediated signaling, may contribute to Alzheimer's disease (AD) pathogenesis. However, the nanoscale structures underlying these processes remain difficult to resolve in situ by conventional fluorescence microscopy due to the diffraction limit of light.

Methods. An AAV vector carrying HA-TurboID-hCD9 under the astrocyte-specific GFAP promoter was injected into the hippocampal dentate gyrus of 3-month-old mice, whose brains were collected, fixed, and sectioned three weeks post-injection. These fixed sections then underwent expansion microscopy (ExM), which embeds tissue in a swell-able polymer and isotropically enlarges it, effectively increasing its spatial resolution. Expanded sections were then acquired as single-plane and z-stack confocal images, focusing on regions of interest including synaptic regions immunolabeled for VGLUT1 and Homer-1 as pre- and postsynaptic markers, and HA-CD9-positive structures representing astrocytic membranes. Three-dimensional Imaris reconstructions were generated to better examine their special relationships within surrounding neuronal architecture.

Results. ExM achieved a 7-8x tissue expansion, corresponding to an effective lateral resolution of around 25-30 nm compared with ~200 nm using conventional fluorescence microscopy, allowing for the visualization of EV-sized structures (~50–150 nm). Fine astrocytic processes and TurboID-CD9-positive structures could be visualized as structures with defined shapes and volumes rather than point-like signals, facilitating assessment of their three-dimensional spatial relationships within surrounding neuronal architecture. ExM improved the spatial resolution of VGLUT1 and Homer-1 labeling, allowing pre- and postsynaptic structures to be clearly resolved in close apposition. Spherical HA–TurboID–CD9-positive structures consistent with EV-like morphology were resolved in neuronal structures, including signals localized beyond clearly identifiable astrocytic processes.

Conclusions. ExM enables high-resolution, three-dimensional visualization of fine astrocytic and HA-positive structures in brain tissue using conventional fluorescence microscopy without specialized equipment. Combined with astrocyte-selective TurboID-CD9 labeling, this approach provides a framework for investigating the distribution and fate of astrocyte-derived EVs in the brain.

Keywords: Alzheimer's Disease, expansion microscopy, astrocytes, extracellular vesicles, high-resolution imaging

Cite as: Matveev S, Akamine S, McQuaker-Kajiwara J, et al. High Resolution In Situ Visualization of Astrocyte-Associated Nanoscale Structures Through Combined TurboID Labeling and Expansion Microscopy. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-57.

P-60 POSTER Poster session II Surgery, Wound Care & Orthopedics

Retrospective Review of Upper Extremity Acral and Nonacral Glomus Tumors

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Presenter: Taylor Rakauskas, Department of Orthopedic Surgery

Introduction. Glomus tumors are benign mesenchymal tumors composed of modified smooth muscle cells of the glomus body. Acral glomus tumors are commonly found in the upper extremities, specifically in the subungual area of the digits. Upper extremity nonacral glomus tumors are considerably less common and remain inadequately explored in the literature. The aim of this study was to further characterize and compare the clinical presentation and treatment outcomes for upper extremity acral and nonacral glomus tumors.

Methods. A retrospective review was performed on patients diagnosed with upper extremity glomus tumors across multiple tertiary referral centers from 1994-2026. Patients were categorized as having acral or nonacral tumors. Primary outcomes of interest included demographic variables, tumor size, tumor location, treatment pathways, and history of recurrence.

Results. A total of 159 patients were identified, including 98 acral (61.6%) and 61 nonacral (38.4%). Acral tumors were more common in women (65.3%), whereas nonacral tumors predominated in men (82.0%; p,0.001). Acral lesions most frequently involved the ring finger (25.5%), middle finger (20.4%), and thumb (16.3%) while nonacral tumors were most common in the forearm (44.3%), arm (19.7%), and elbow (16.4%) (all p<0.001). Surgical excision was performed in 90.8% of acral and 65.6% of nonacral cases, with recurrence rates of 15.7% and 32.5%, respectively (p=0.038).

Conclusions. This study represents one of the largest analyses of acral and nonacral upper extremity glomus tumors. Acral tumors are more likely to be found in patients with a history of neurofibromatosis, while nonacral tumors were significantly more common in males. While age at diagnosis was similar between groups, recurrence rates following surgical excision were significantly higher in nonacral tumors.

Keywords: glomus tumor, upper extremity, hand

Cite as: Rakauskas T, Tarrat Guadalupe C, Sherman C, Aziz K. Retrospective Review of Upper Extremity Acral and Nonacral Glomus Tumors. Presented at: Mayo Clinic Research Fellows' Association (MRFA) Research Day; October 29, 2026; Jacksonville, FL. Abstract P-60.

P-62 POSTER Poster session II Neuro-oncology

Distinct circadian rhythms in the glioblastoma margin and core regulate time-dependent temozolomide sensitivity

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Presenting author underlined. Principal investigator: Hugo Guerrero-Cazares.

Background. Glioblastoma (GBM) is the most common and lethal primary brain tumor in adults, with a median survival of 15 months despite surgery, radiation, and chemotherapy with temozolomide (TMZ). These devastating trends highlight the importance of understanding the intrinsic mechanisms driving aggressive GBM progression to develop better therapeutic strategies. Recent evidence suggests that GBM tumors have circadian rhythms that regulate the response to chemotherapy with TMZ, with maximum efficacy obtained when administered in the morning compared to evening. However, we do not know if daily rhythms in clock gene expression and sensitivity to TMZ vary depending on GBM location.

Hypothesis. Because GBM biology can significantly change depending on where cells are located within a tumor and the surrounding tumor microenvironment, we hypothesize that circadian regulation of clock gene expression and sensitivity to daily TMZ varies depending on location.

Methods. To test this hypothesis, we obtained GBM cells from tumor margin and core regions and transduced them with luciferase reporters driven by the promoters of the core clock genes BMAL1 or PER2. We next recorded clock gene-driven bioluminescence continuously for 3 days to measure circadian oscillations in BMAL1 and PER2 expression. We found that GBM cells from the tumor margin show daily rhythms in clock gene expression, while cells from the tumor core show a disrupted circadian clock. To assess if daily sensitivity to TMZ also varied between margin and core regions of GBM, we treated cells with various TMZ doses at two circadian times and measured viability 72h post-treatment using the AlamarBlue assay.

Results. We found that GBM margin, but not core cells, show time-of-day differences in sensitivity to TMZ. These results suggest that circadian regulation of clock gene expression and daily TMZ efficacy depends on the spatial origin of the cells within the tumor.

Conclusion. Ultimately, these results can lead to personalized, time- and location-informed strategies that improve GBM outcomes.

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P-63 POSTER Poster session II Women's Health, Weight & Shared Decisions

Breastfeeding Beyond the Clinic: The Value of Virtual Lactation Consultation

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Introduction. Virtual lactation consultants have been assisting mothers in the first year postpartum to help reduce anxiety through easily accessible evidence-based virtual support. Virtual lactation companies have Certified International Board of Lactation Consultant Examiners (IBLCE) on staff to provide medical and clinical expertise focusing on patient-centered care.

Longitudinal evaluation is needed to determine whether ongoing virtual support helps mothers maintain positive breastfeeding experiences and overcome feeding challenges.

Methods. This prospective, single-arm pilot study enrolled lactating mothers receiving care through Mayo Clinic Family Medicine. Eligible participants intended to breastfeed or express breastmilk, provided informed consent, and had access to an internet-enabled device. Breastfeeding experience was evaluated using the Maternal Breastfeeding Evaluation Scale (MBES).

Results. Twenty-four participants completed the initial assessment, and 22 completed the 3-month assessment, corresponding to a 91.7% follow-up rate.

54% of the babies enrolled were over 12 weeks old. 88% of the mothers agreed initially that being able to nurse was important, although 24% of them did not enjoy nursing. 50% worried about initial weight gain and 42% found breastfeeding emotionally draining.

The average across the 10 analyzed items increased from 3.60 ± 0.65 at the initial assessment to 3.66 ± 0.62 at 3 months. Seven of the 10 items showed numerical improvement, but no individual item had a statistically significant change.

Conclusions. Participants generally entered the study with positive breastfeeding experiences, yet many still reported meaningful concerns or challenges. This suggests virtual lactation support may be relevant not only for mothers having trouble, but those seeking reassurance or continued guidance.

Continued participation across later postpartum stages further suggests that lactation support may remain valuable as breastfeeding needs and challenges evolve.

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